

DRAFT FINAL XSA RUNS

Data and Methods

The assessments were conducted using XSA in R using the FLXSA package (part of the FLR-project, Kell et al., 2007; <http://www.flr-project.org/>). The catch-at-size (CAS) data covered the period 1985-2018 and the catch at age (CAA) data were generated using a statistical approach that revealed six age groups.

Four standardized CPUE series extending up to the latest years were considered: Greek longliners (SCRS 2020/021), Moroccan longliners (SCRS 2020/026), Spanish longliners (SCRS 2020/043) and Ligurian longliners (SCRS 2020/027). The standardized CPUE indices were not differentiated by age and were considered to be representative of the 2-4 age-group abundances. Fleet catchability was assumed to be independent of year-class size for all terminal years. Considering recent information on maturity (ICCAT, 2020), it was assumed 15% and 65% maturation at ages 2 and 3 respectively. Full maturity was considered from age 4 onward. Weights-at-age were derived from the mixture analysis and were consistent with the CAA. The F range (fbar) was set to ages 2-4. Natural mortality was assumed constant $M=0.2$

The following configuration was adopted for the XSA control object settings:

shk.n=TRUE, shk.yrs=10, rage=1, maxit=100, shk.f=TRUE, shk.ages=4, fse=0.5, min.nse=0.3, qage=6, window=100, tsrange=20, tspower=1, vpa=TRUE, F range: minfbar=2, maxfbar=4.

Apart from the base run, an additional one assuming discards, based on the scenario developed in the SCRS 2020/028 document, was made.

Results

a. Base run

Model estimates are plotted in **Figures 1-2**. The various diagnostic plots are shown in **Figures 3-11**, while **Figure 12** shows the retrospective plot.

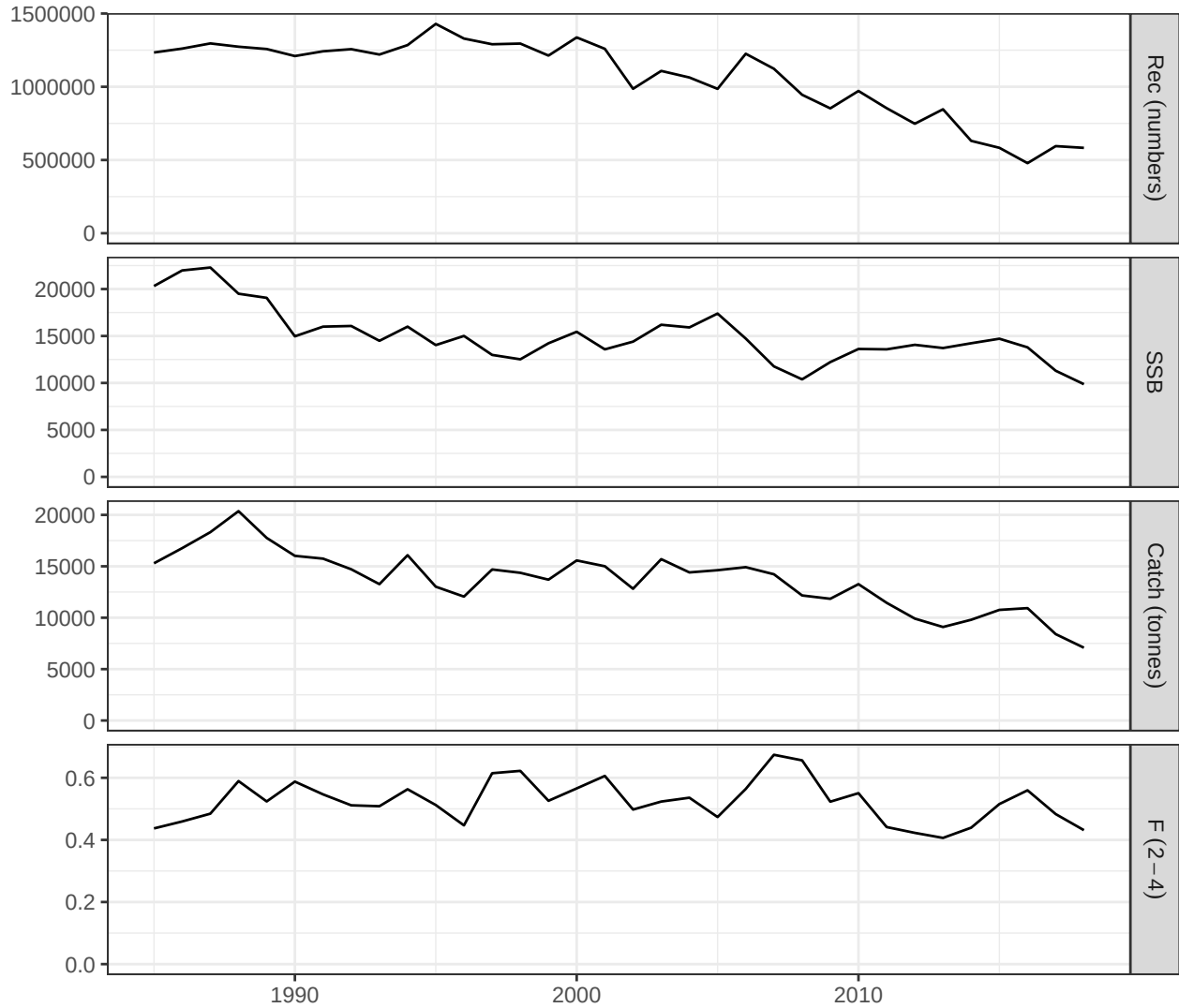


Figure 1. XSA time series estimates for the base run

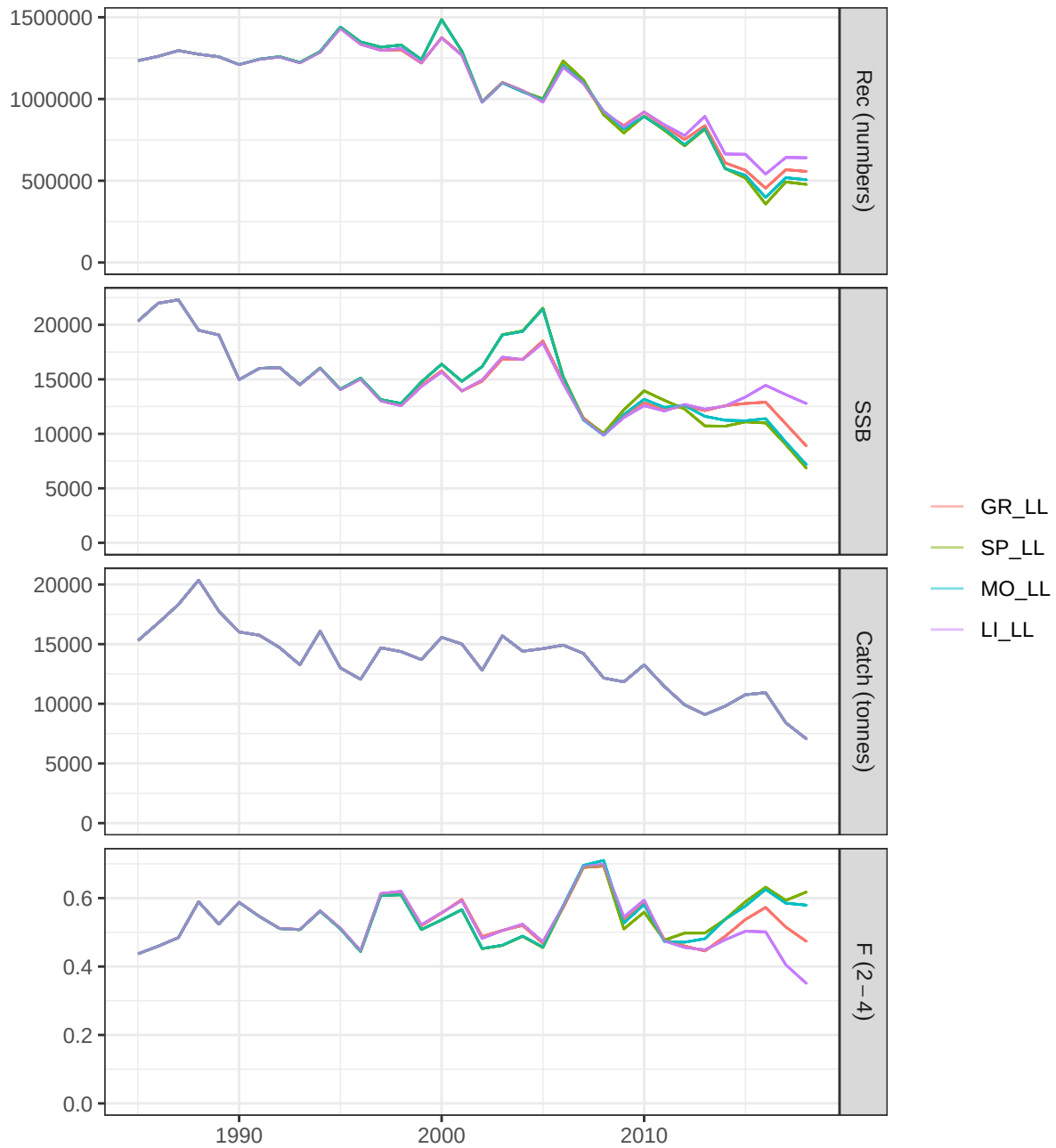


Figure 2. XSA time series estimates by CPUE, for the base run

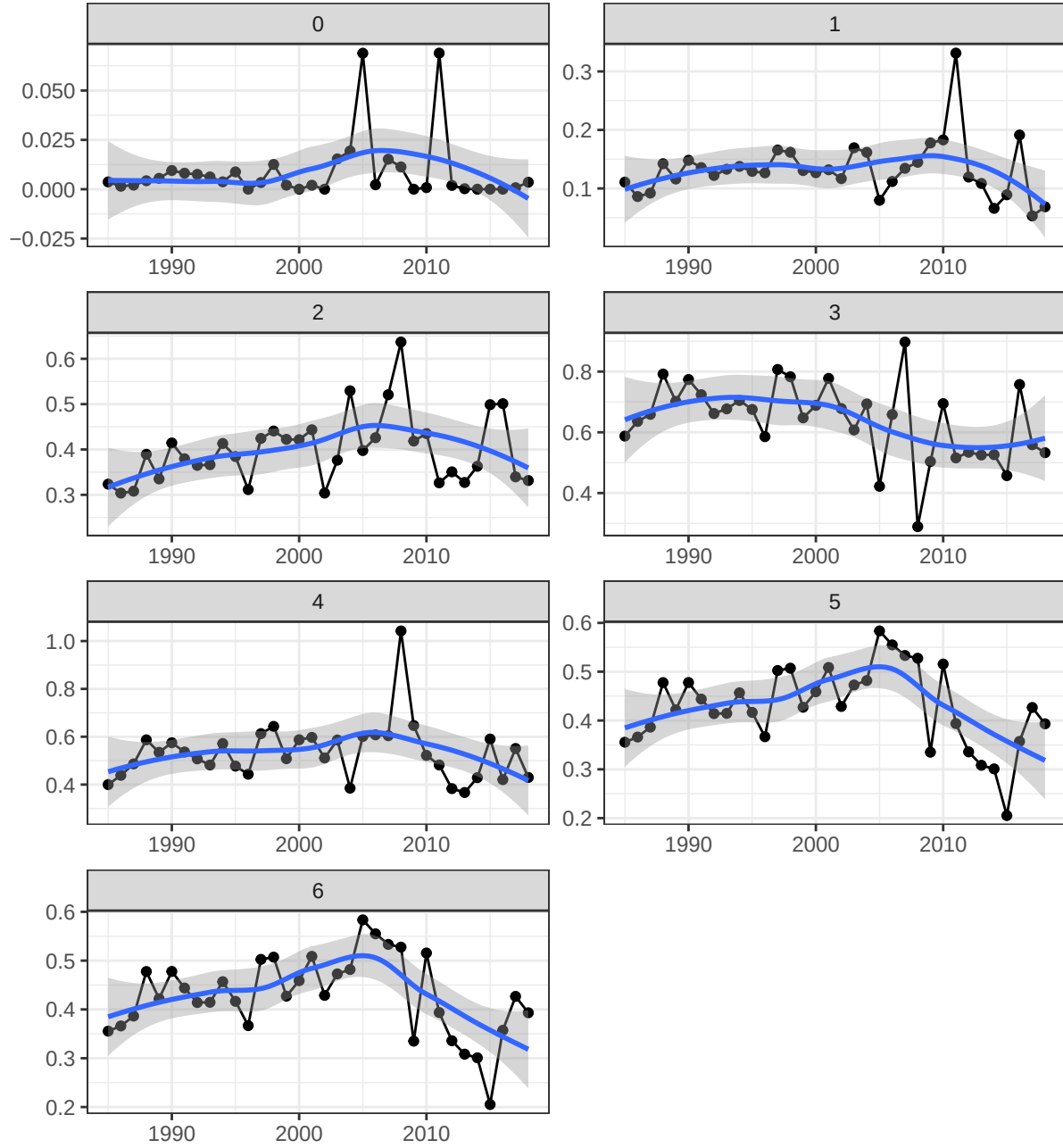
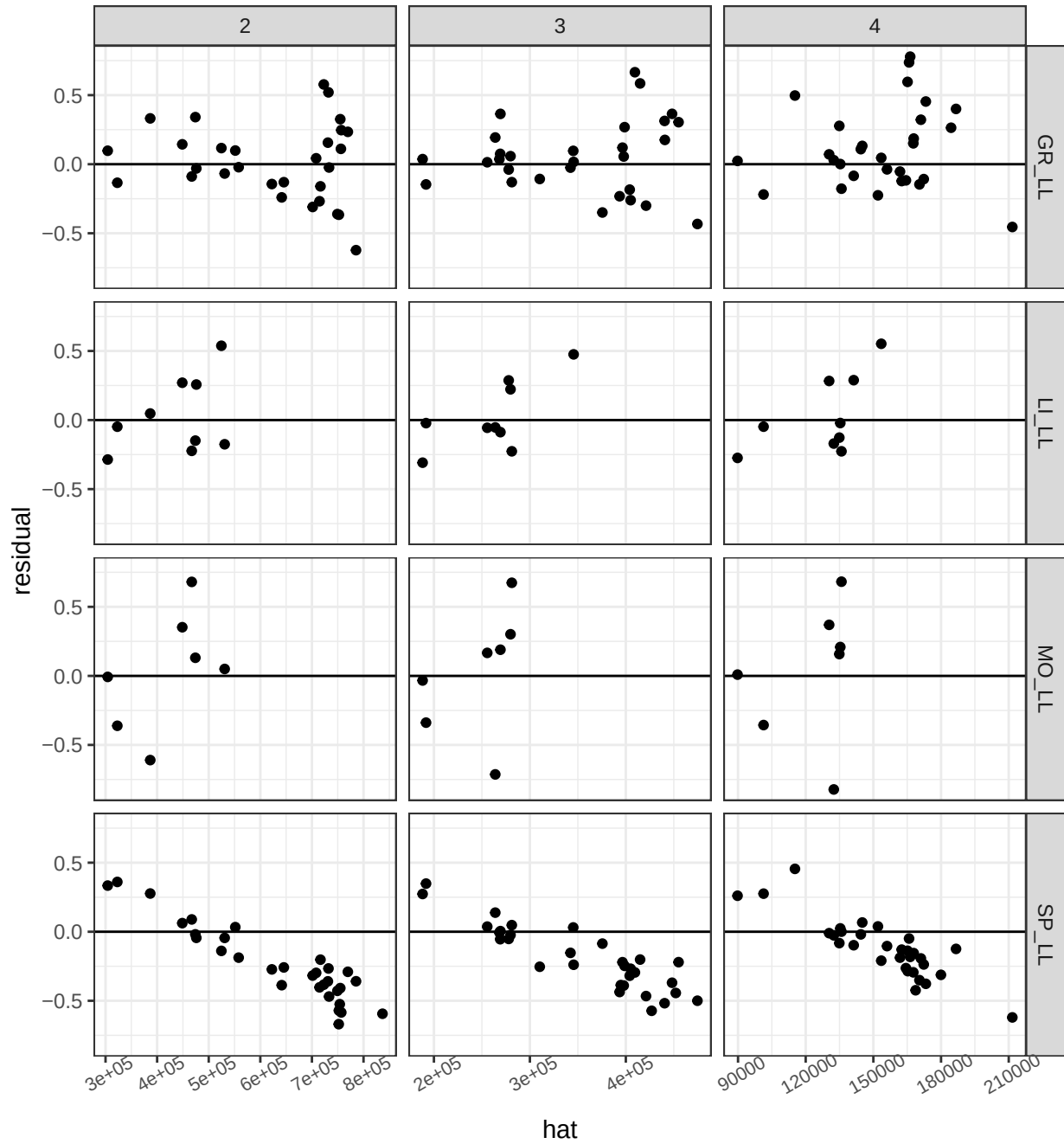


Figure 3. Estimates of harvest-at-age by year for the for the base run; continuous lines represent lowess smoothers.

Diagnostic plots**Figure 4.** Residuals against fitted value for the for the base run.

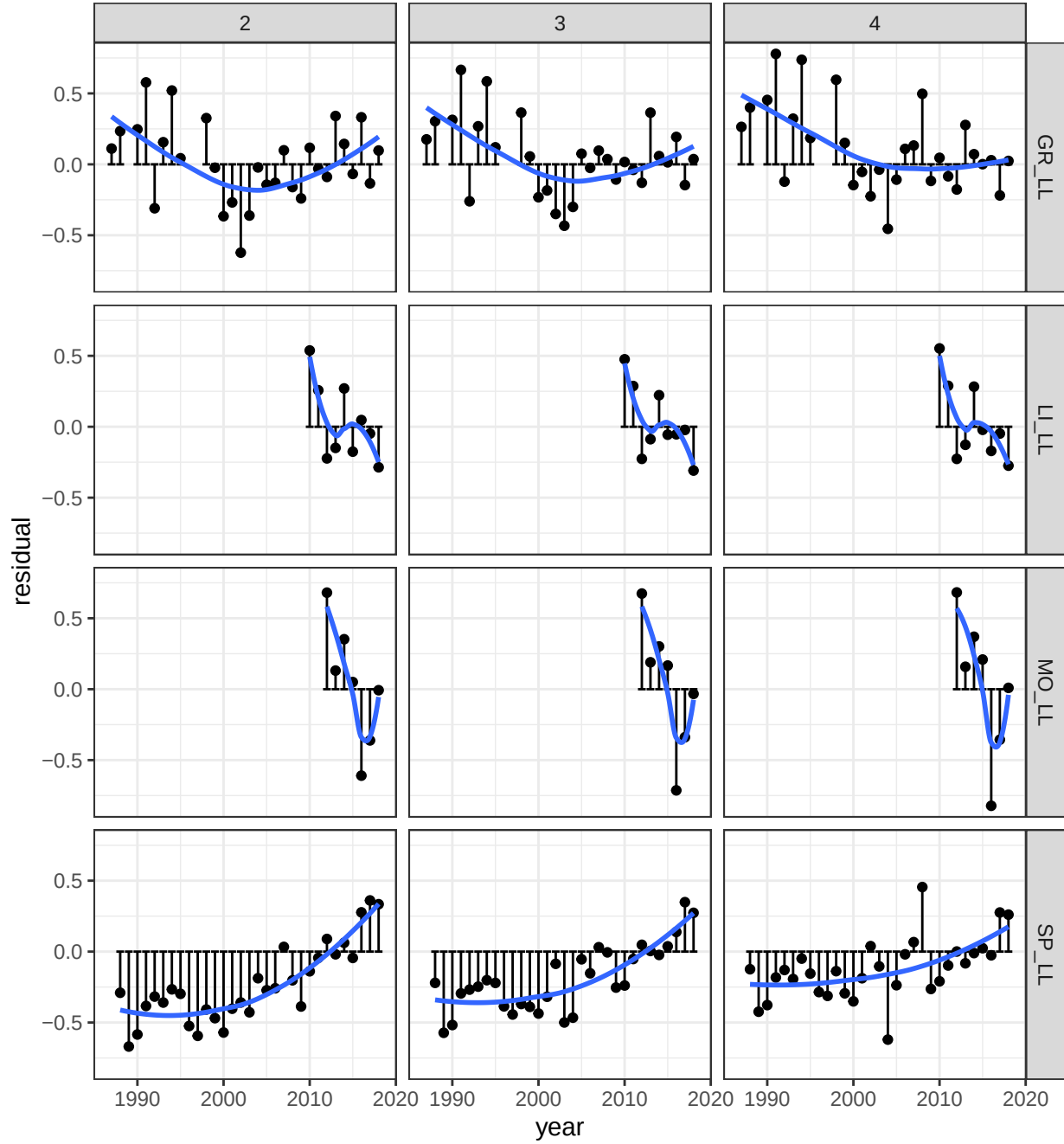


Figure 5. Residuals by fleet against year for the for the base run.

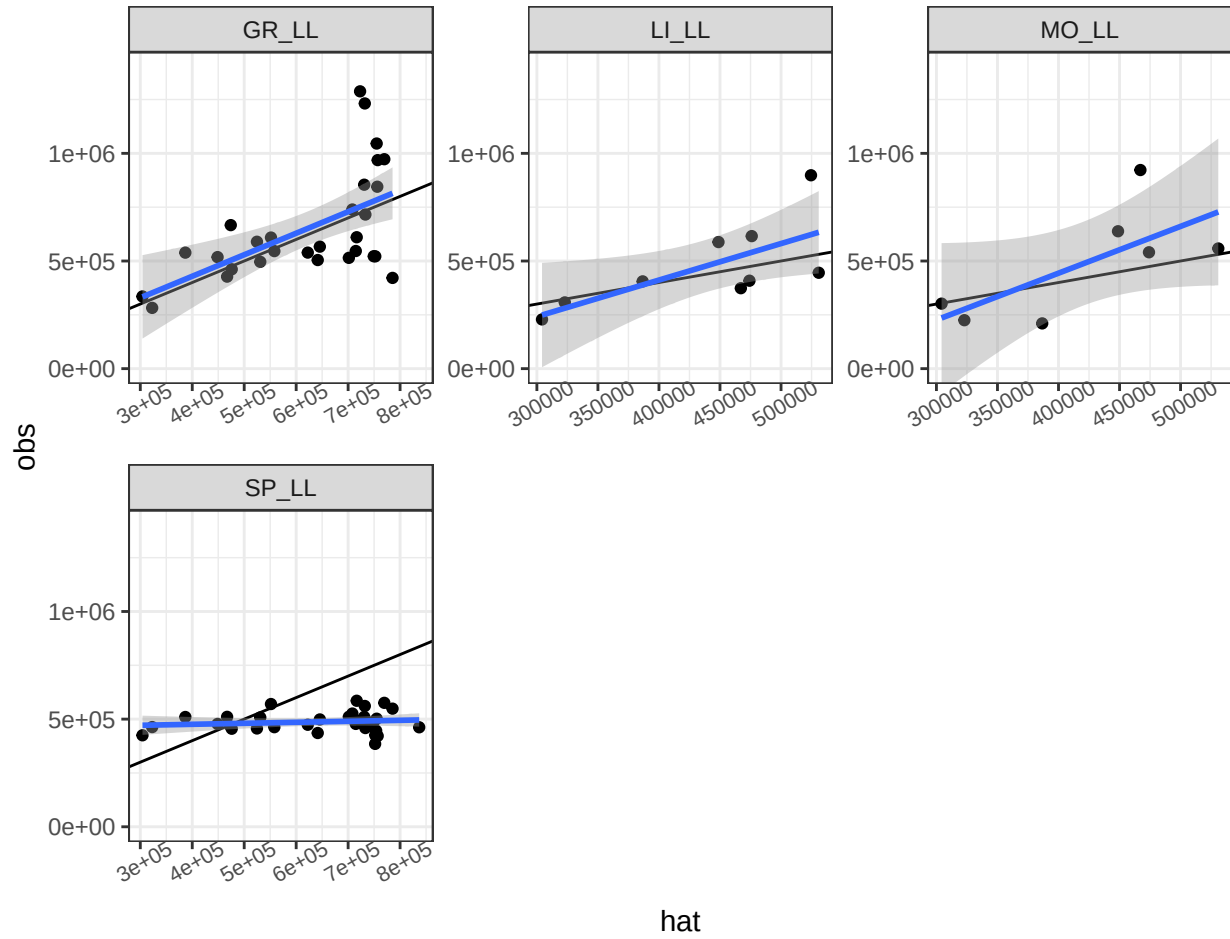


Figure 6. Calibration regression plots for age 2 for the for the base run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

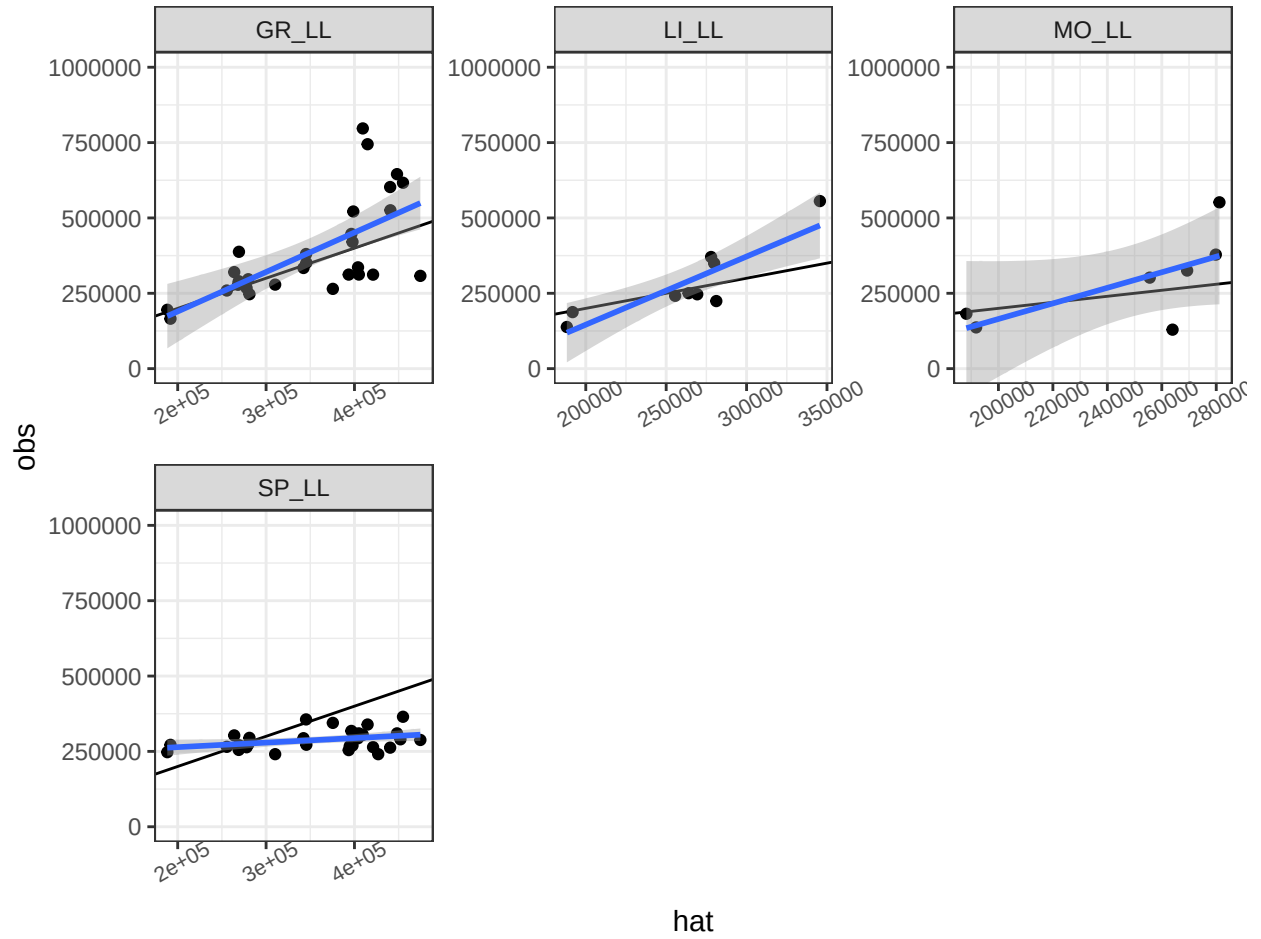


Figure 7. Calibration regression plots for age 3 for the for the base run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

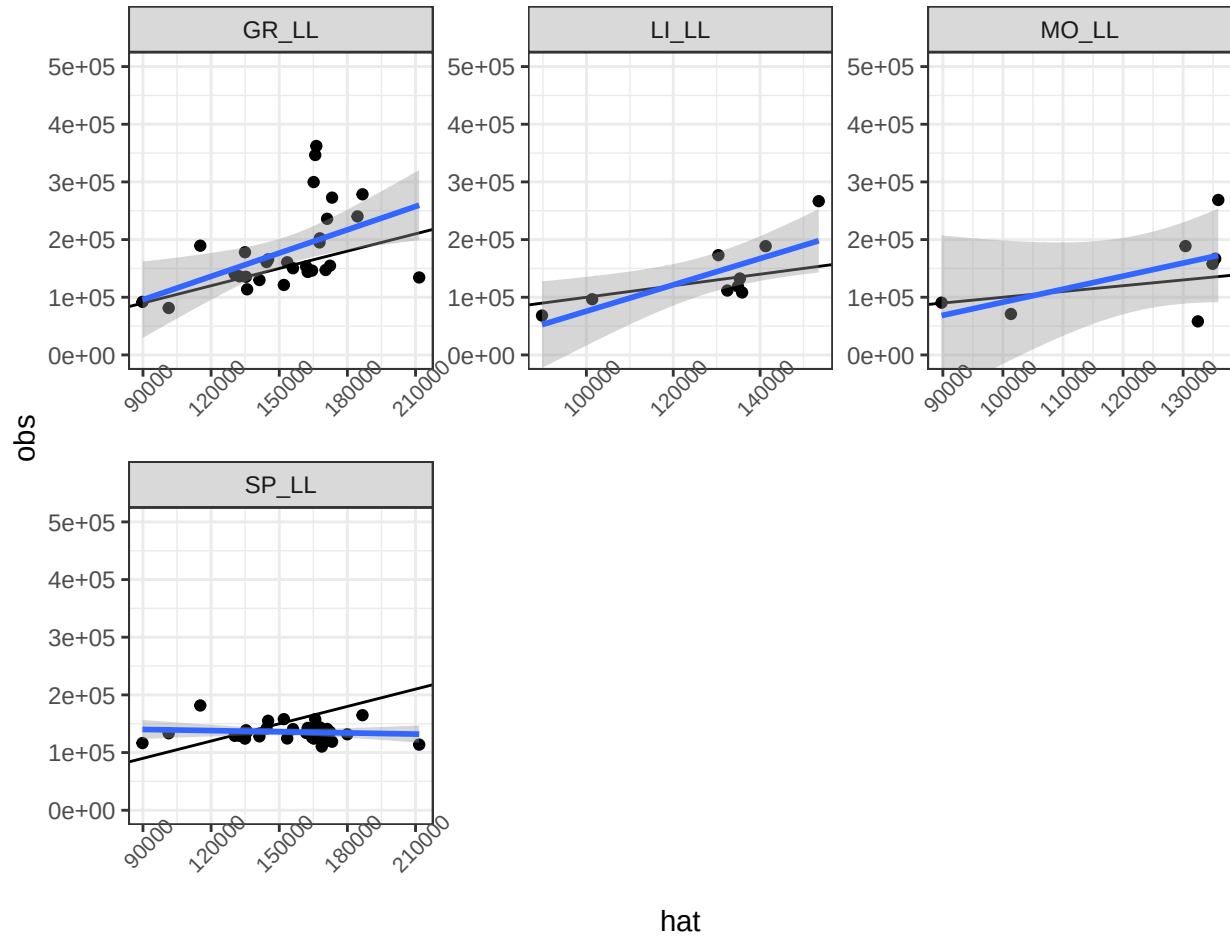


Figure 8. Calibration regression plots for age 4 for the for the base run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

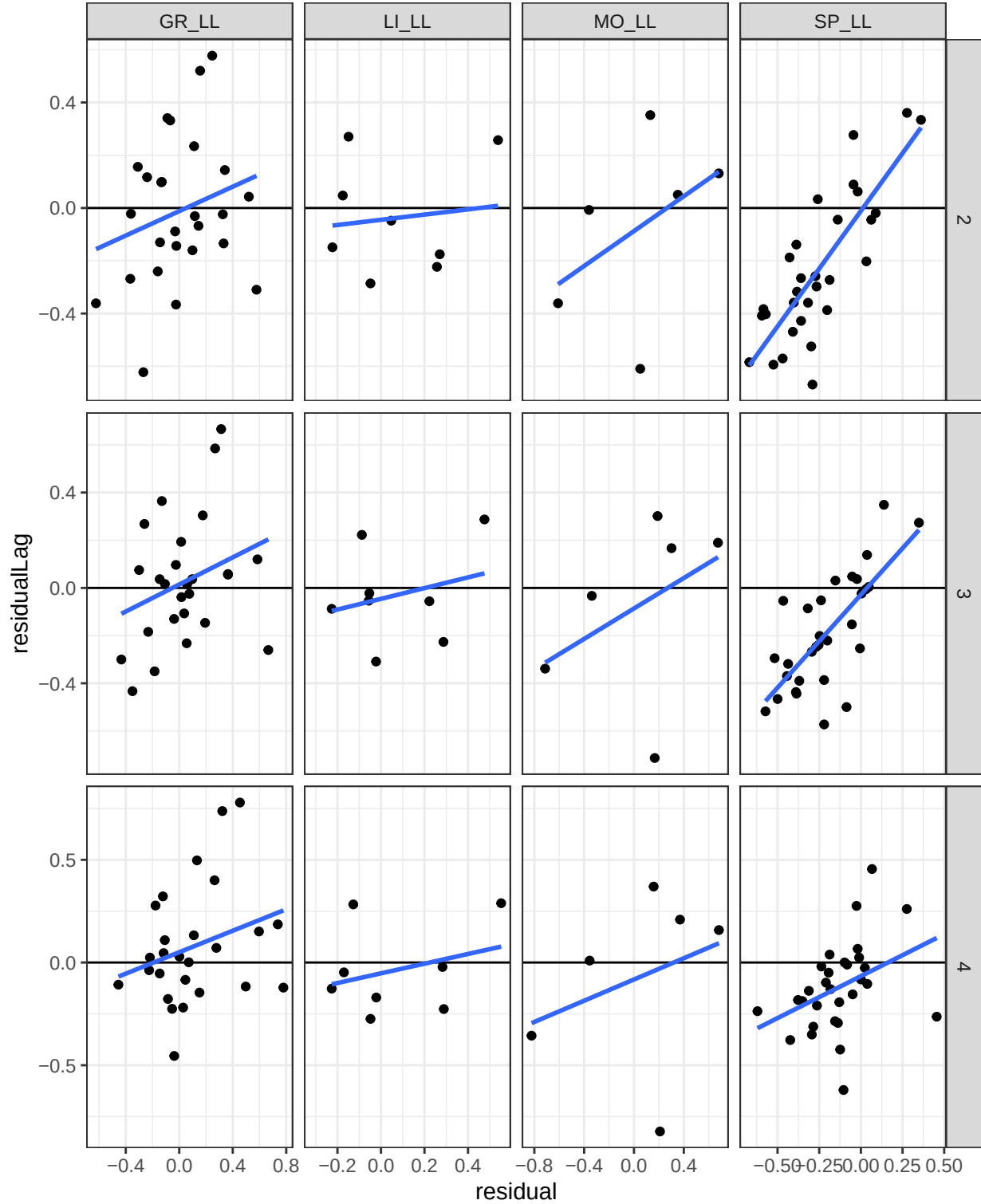


Figure 9. AR plots of lagged residuals for the for the base run. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

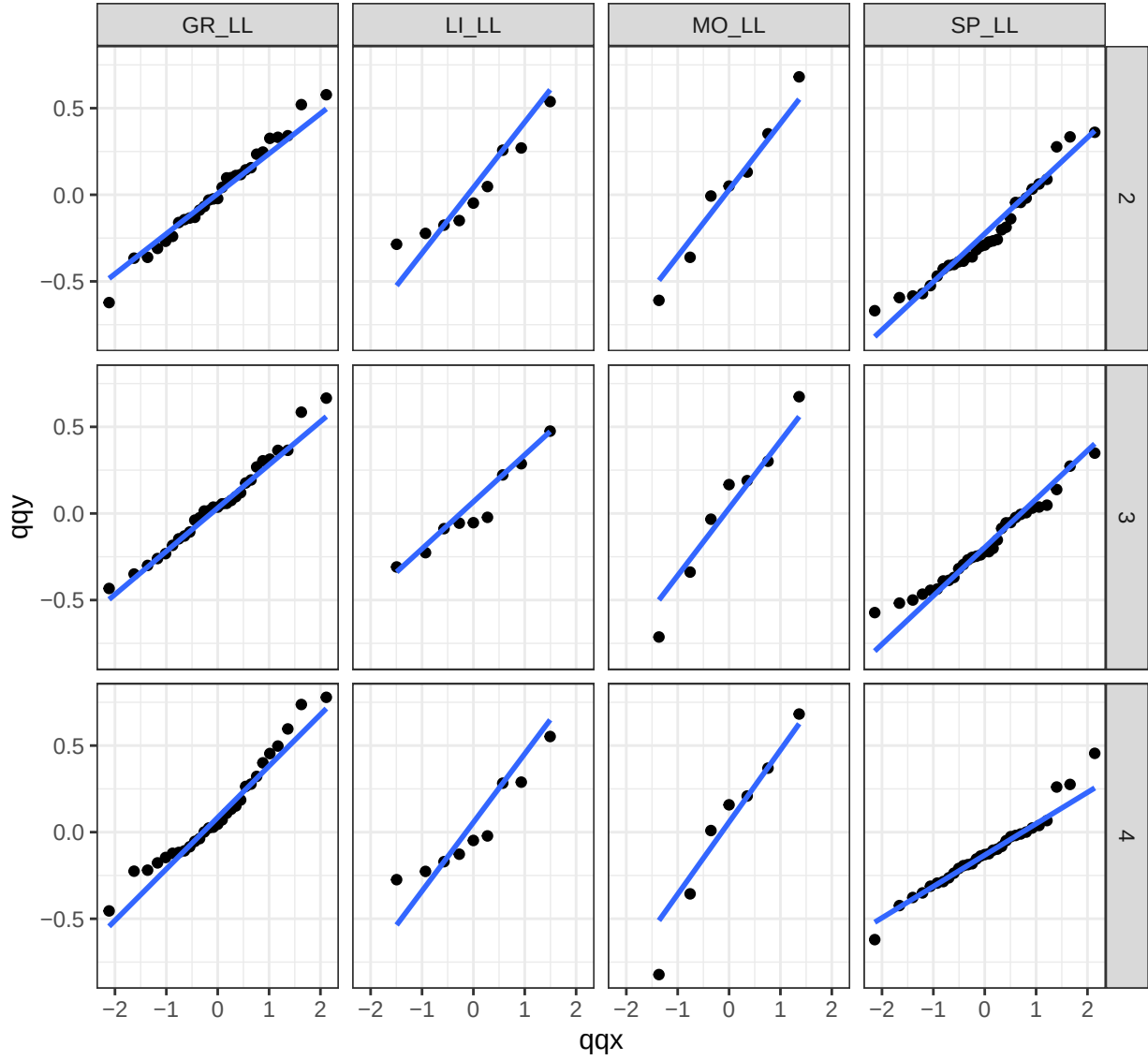


Figure 10. QQ plots to check for log normality (i.e. points and line should coincide) for the for the base run.

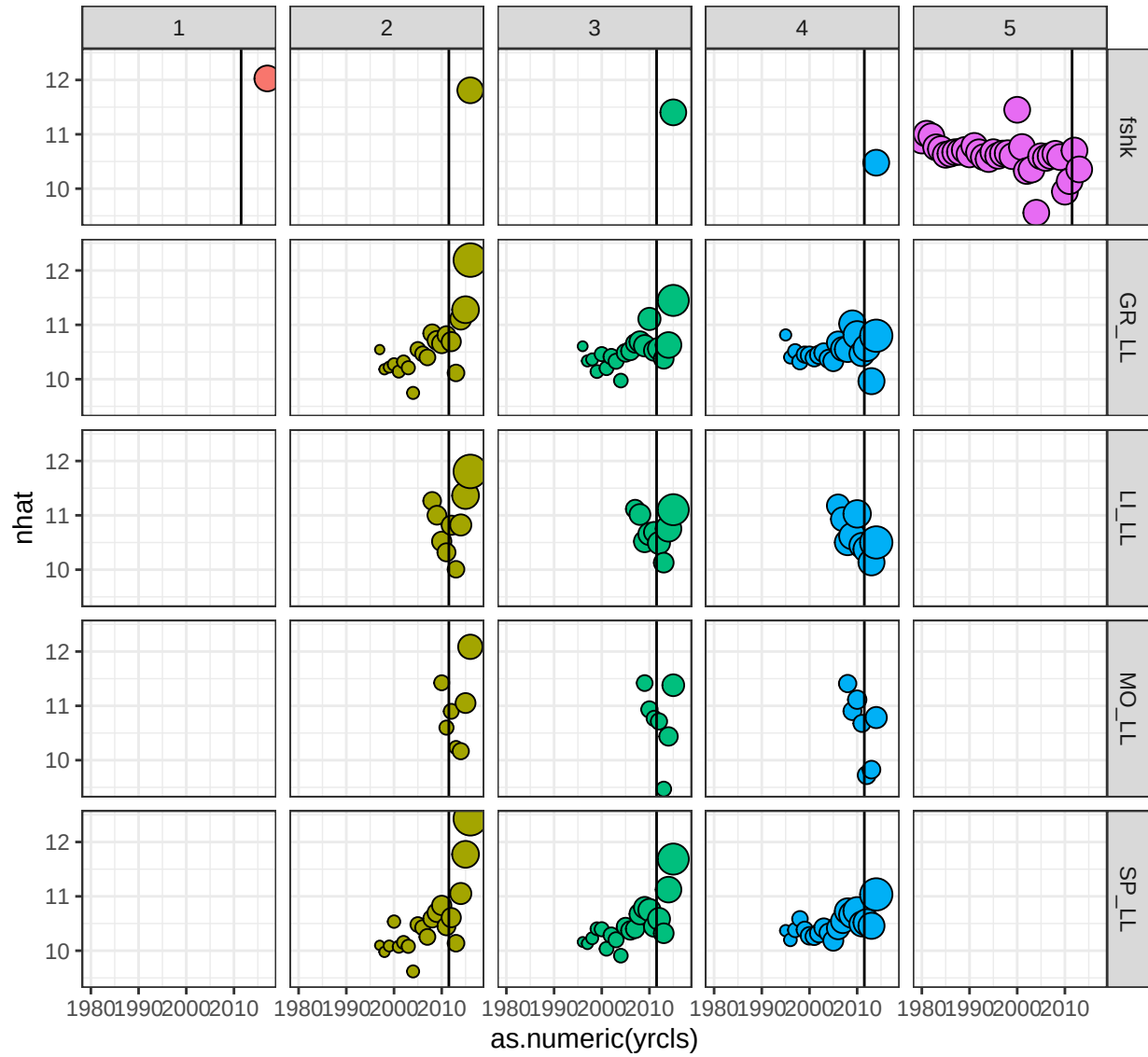


Figure 11. Weights for terminal year Ns for each CPUE observation and shrinkage to the mean $F(fshk)$ for the for the base run. Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

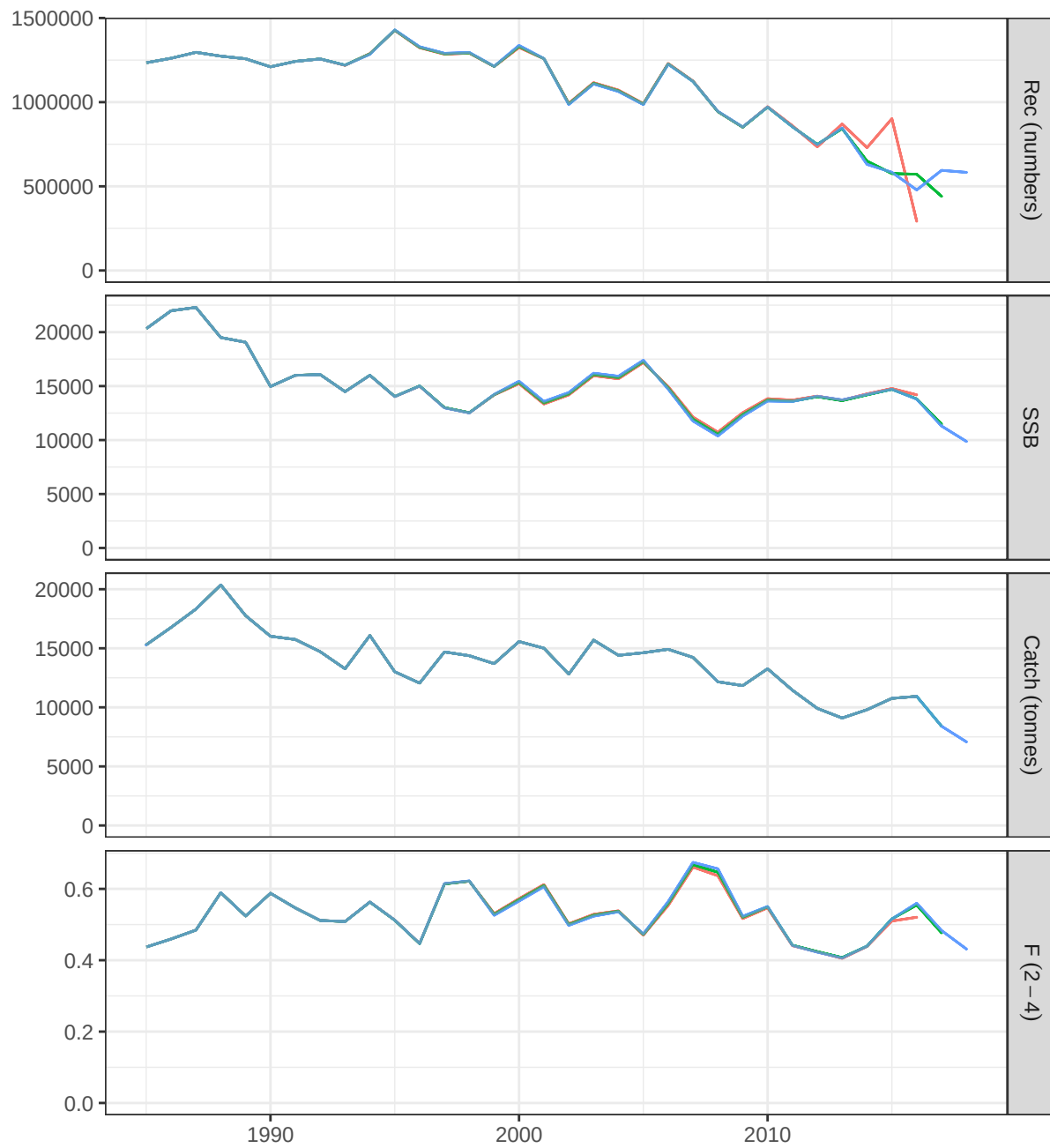


Figure 12. Retrospective plot for the base run.

b. Discard run

Model estimates are plotted in **Figures 13-14**. The various diagnostic plots are shown in **Figures 15-23**, while **Figure 24** shows the retrospective plot.

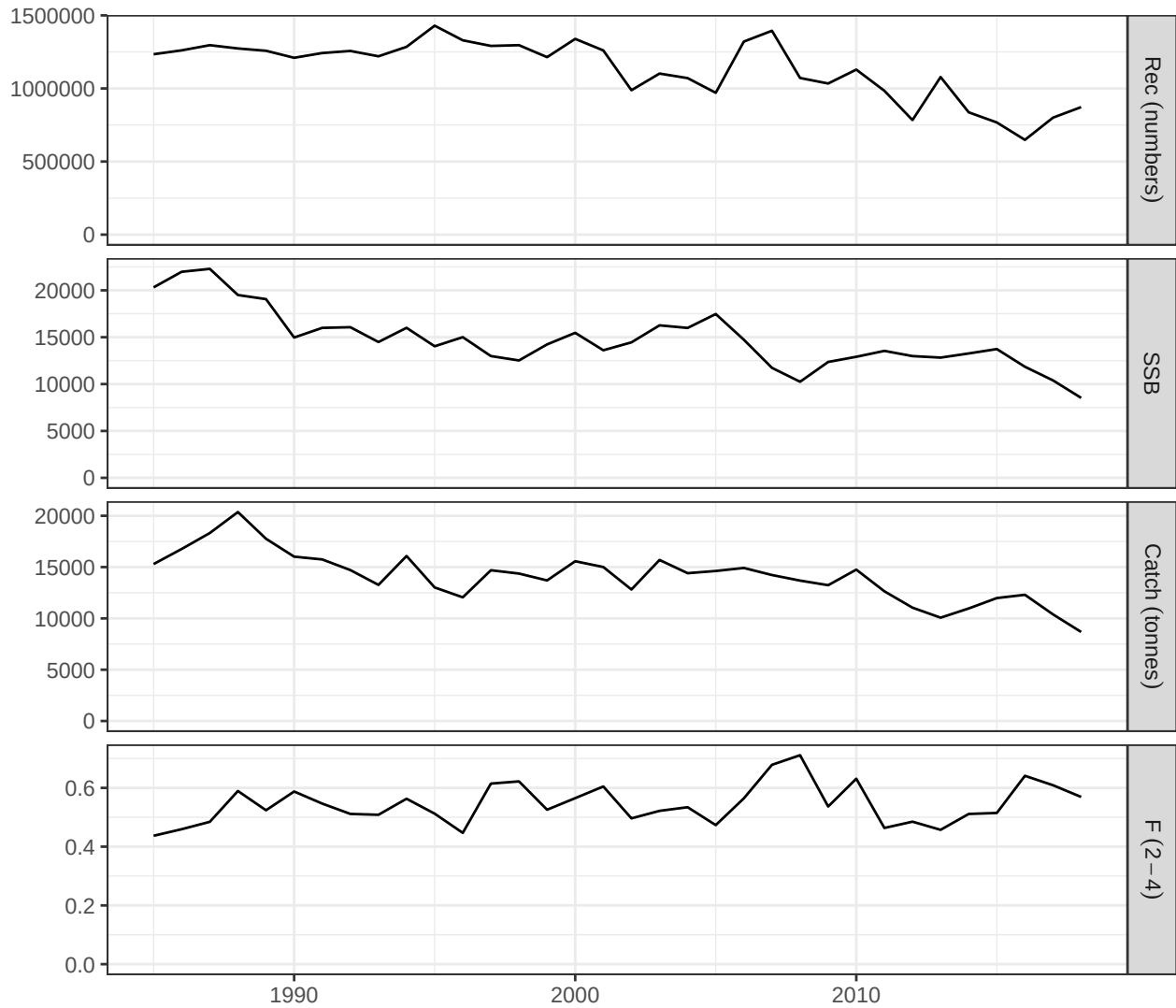


Figure 13. XSA time series estimates for the discard run assuming constant M .

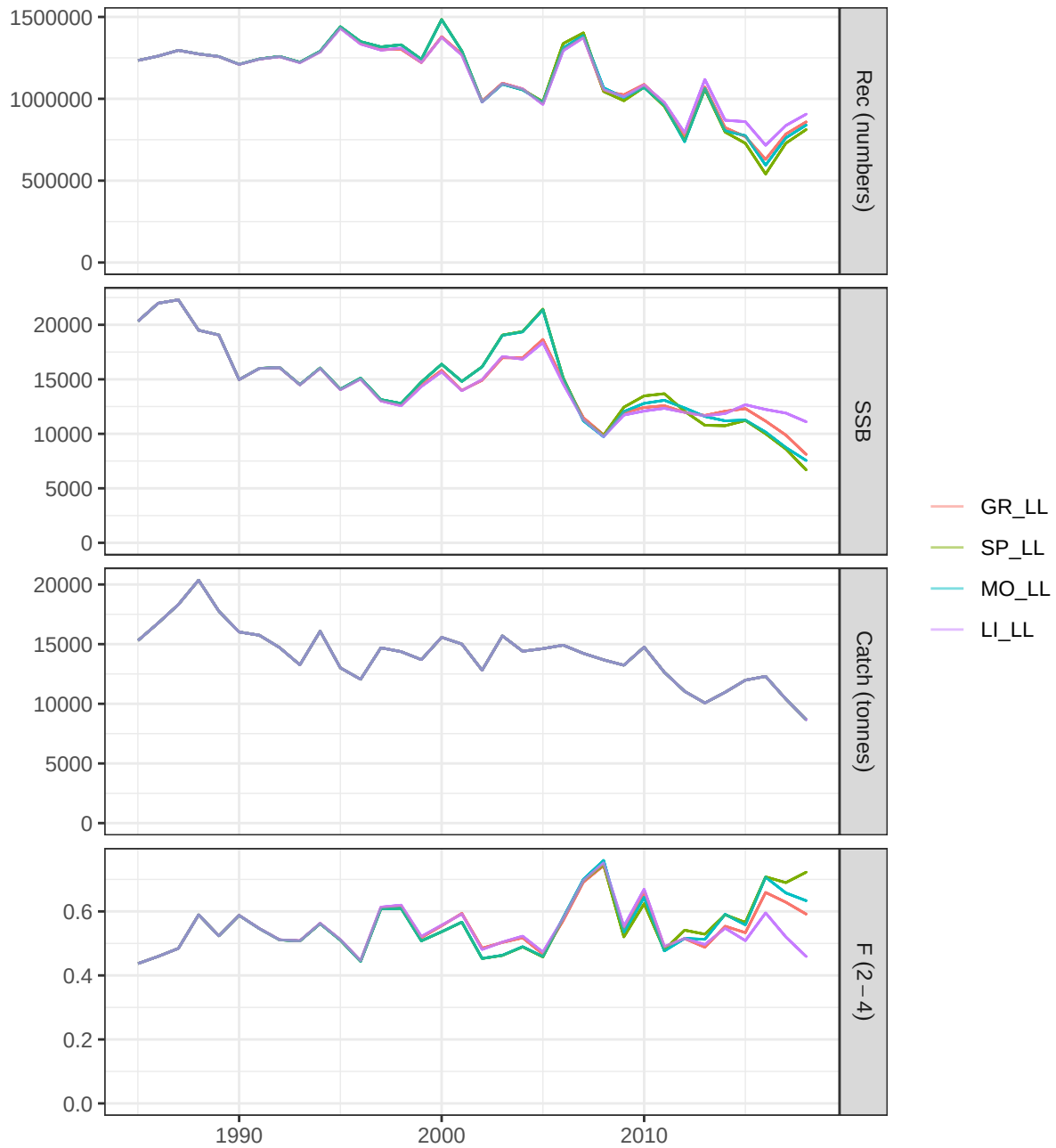


Figure 14. XSA time series estimates by CPUE, for the discard run assuming constant M.

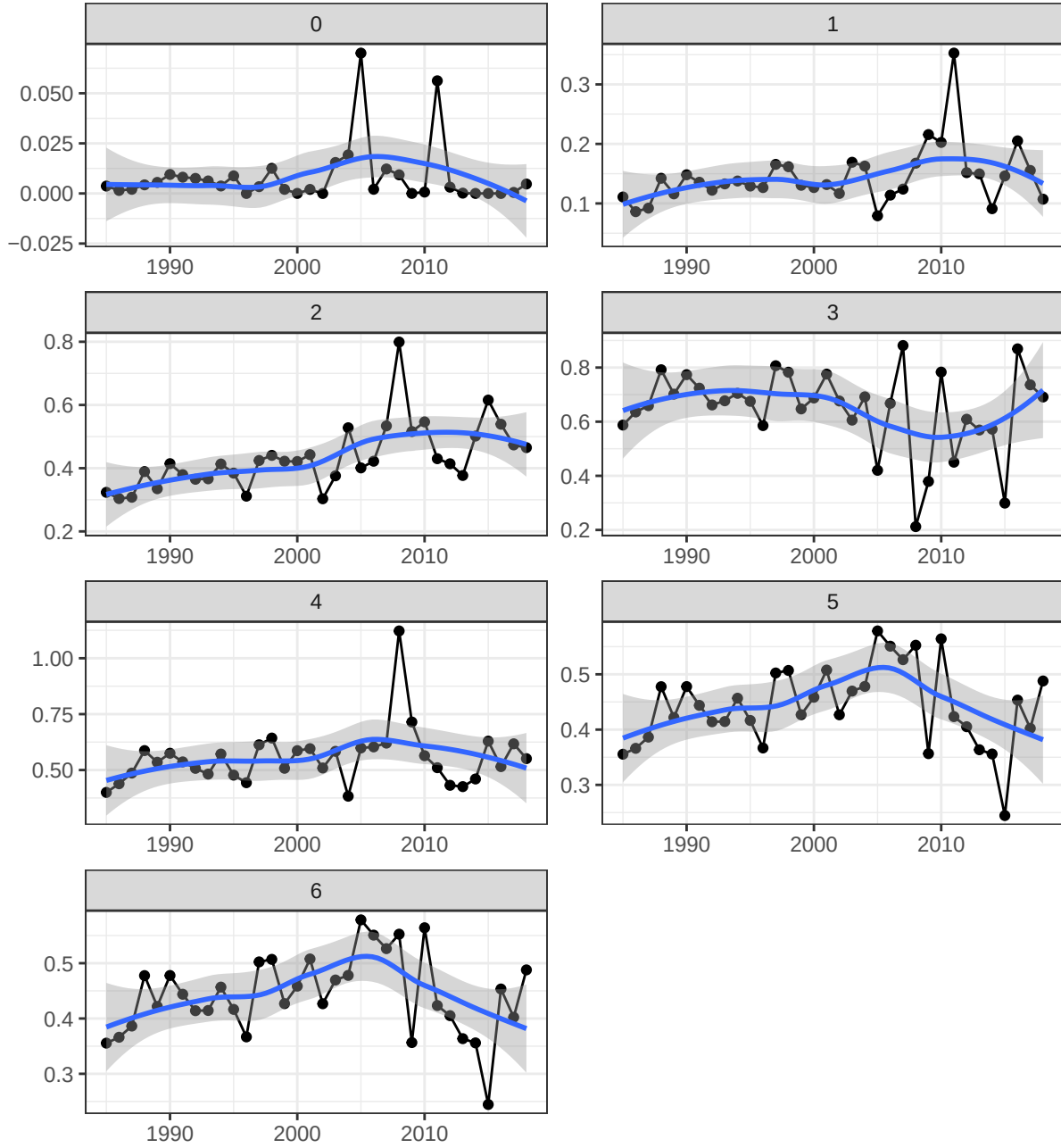
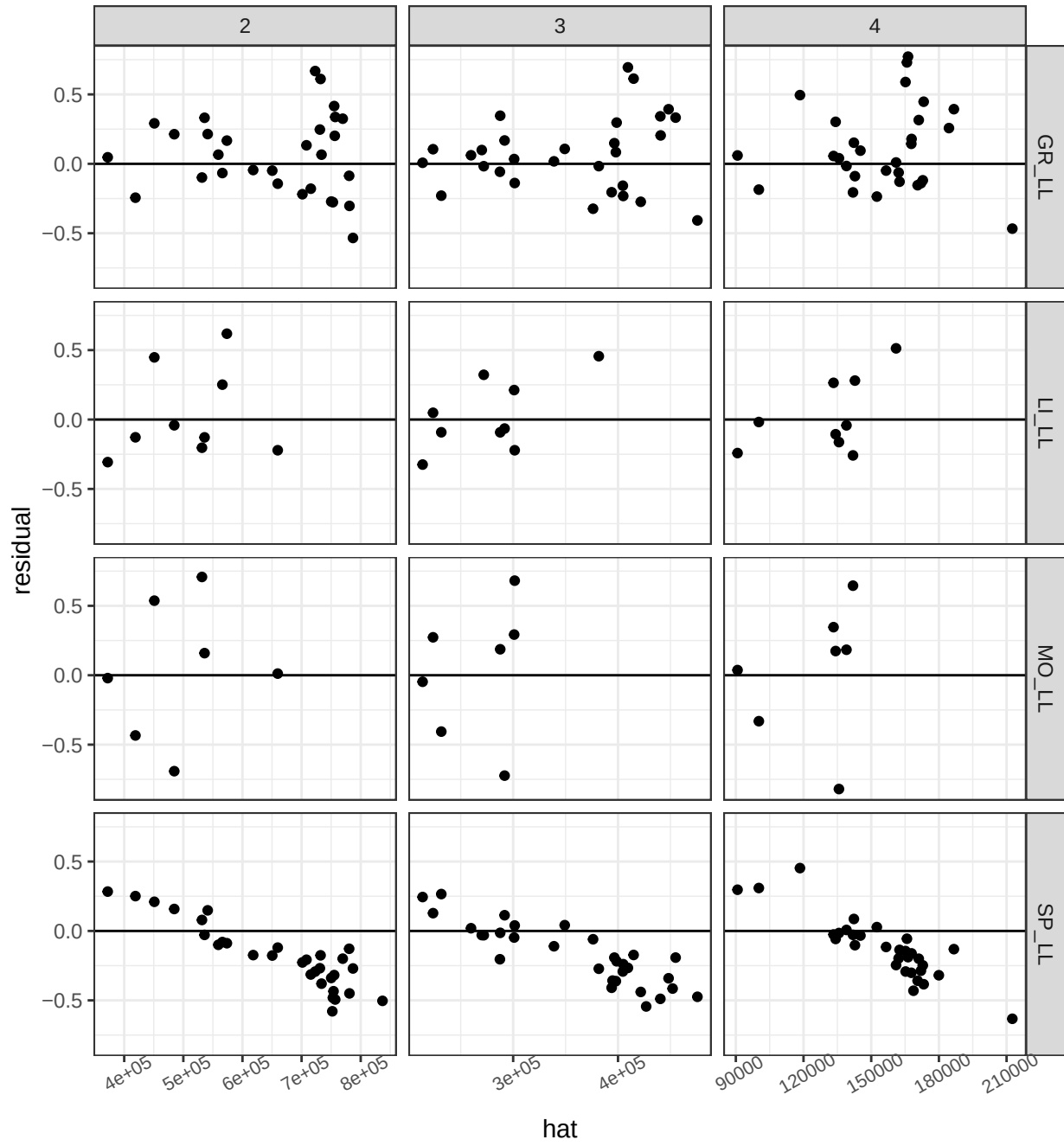


Figure 15. Estimates of harvest-at-age by year for the discard run.

Diagnostic plots**Figure 16.** Residuals against fitted value for the discard run.

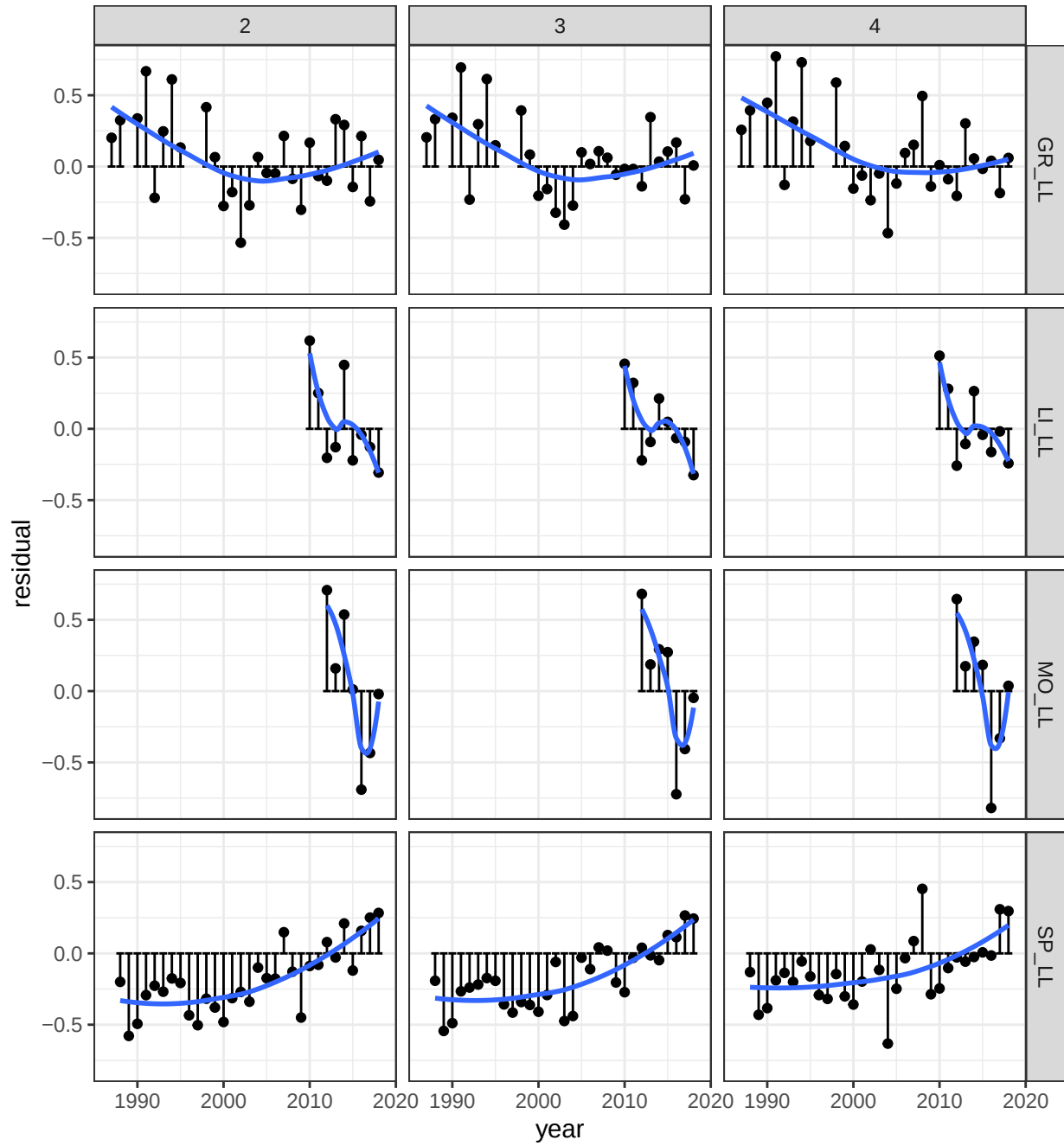


Figure 17. Residuals by fleet against year for the discard run.

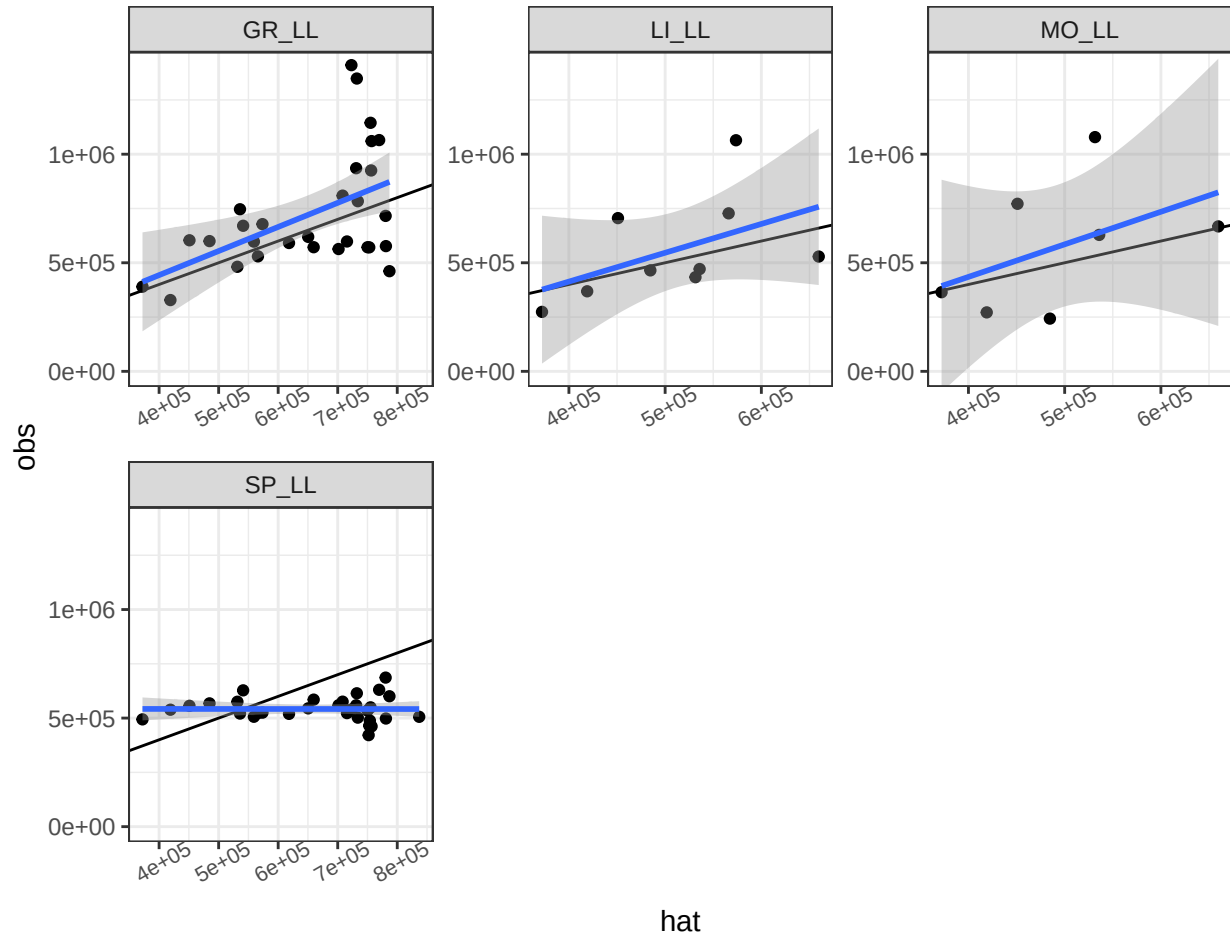


Figure 18. Calibration regression plots for age 2 for the discard run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

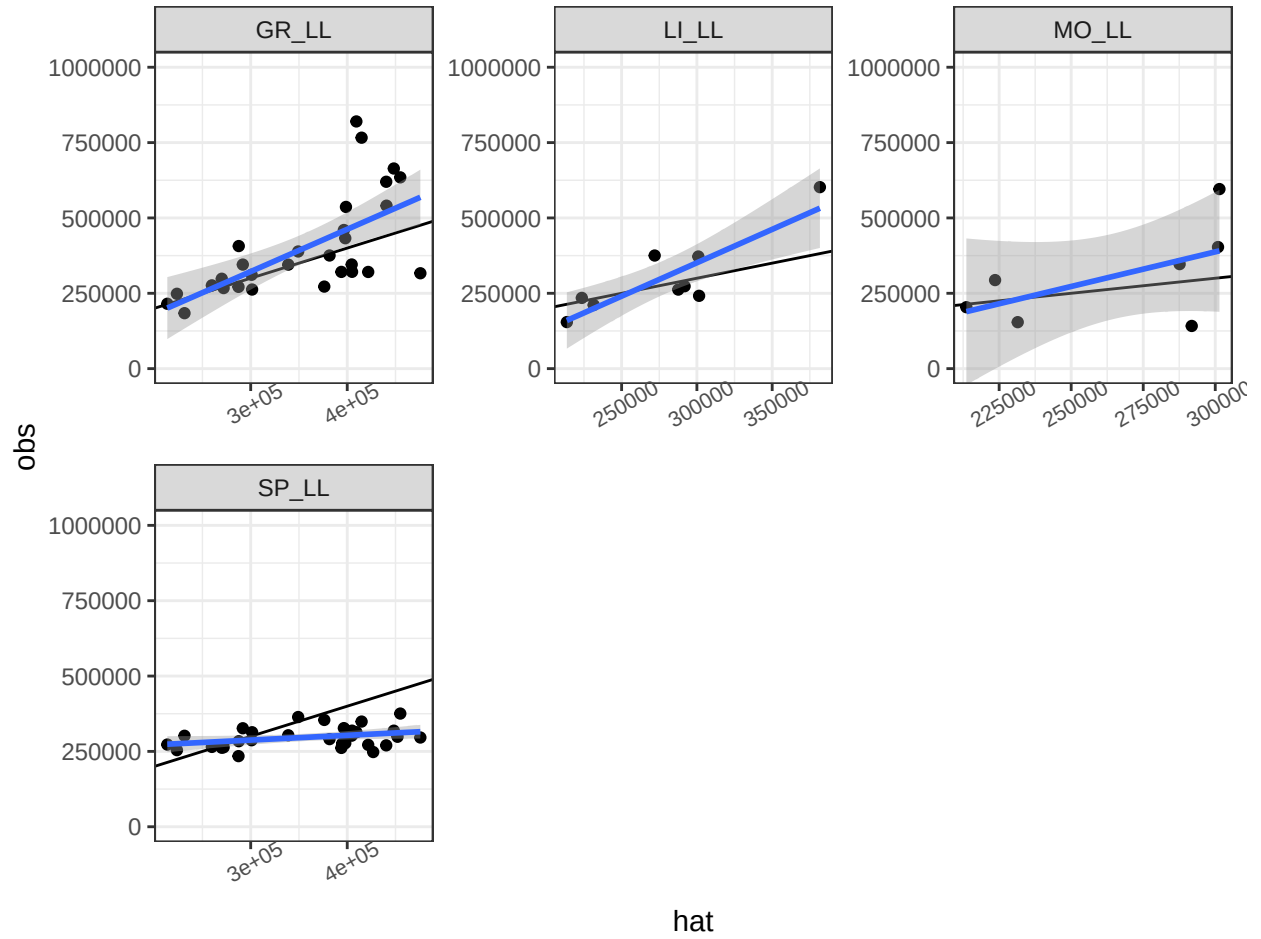


Figure 19. Calibration regression plots for age 3 for the discard run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

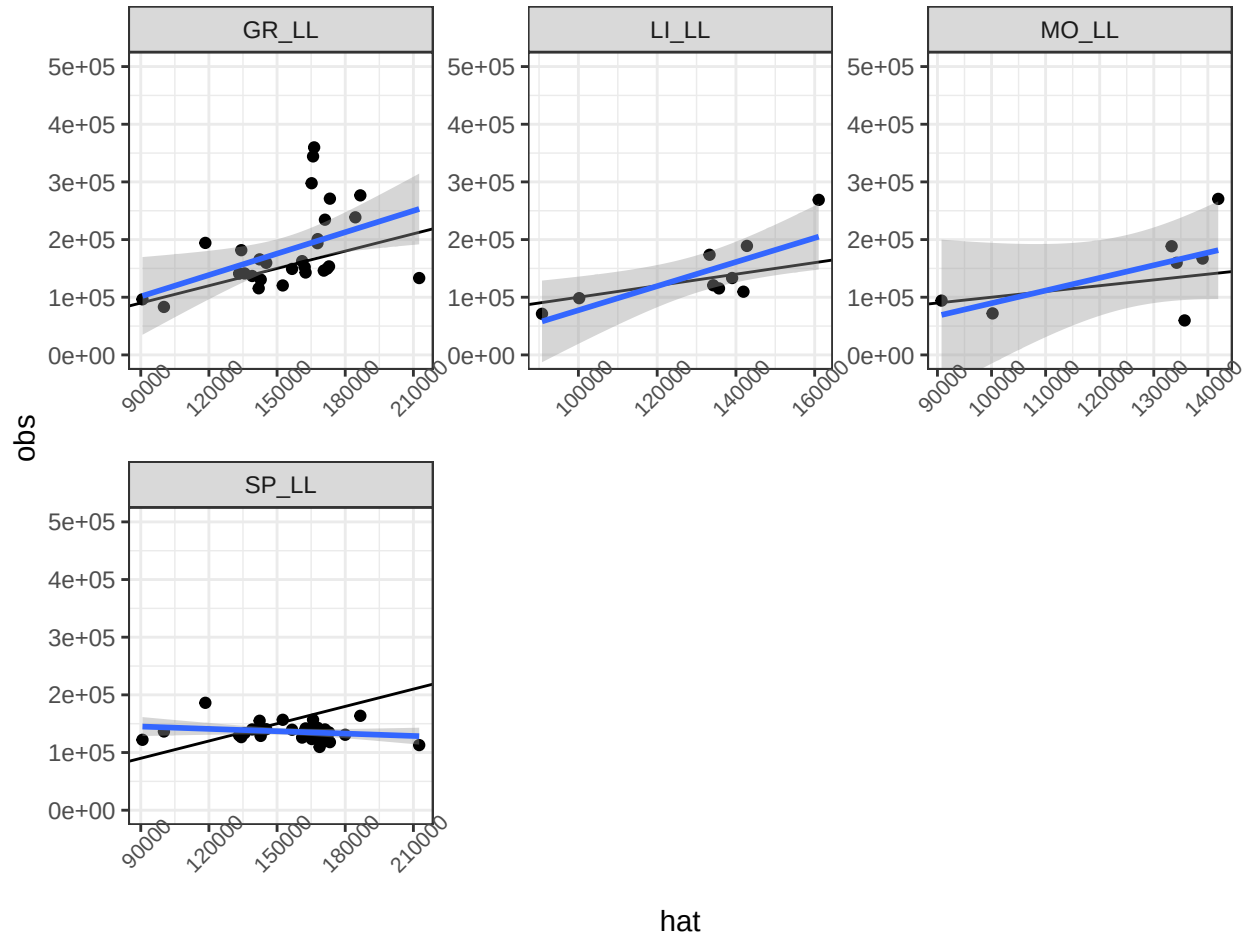


Figure 20. Calibration regression plots for age 4 for the discard run. Blue line is a linear regression fitted to the residual pairs and the black line is the $y=x$ line. If the index is proportional to stock abundance then the blue and black lines should coincide.

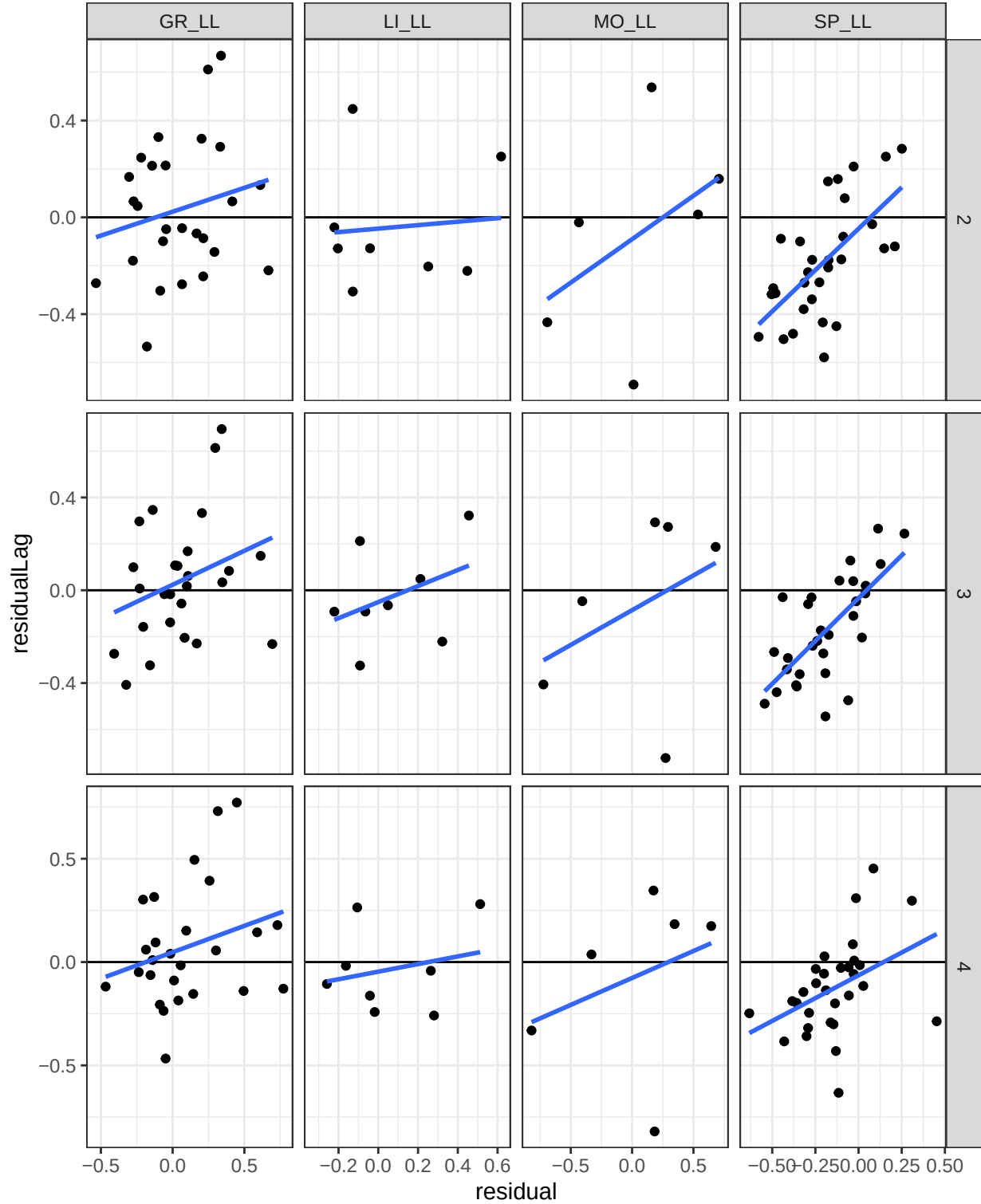


Figure 21. AR plots of lagged residuals for the discard run. Blue line is the regression fitted to the data pairs. If residuals are uncorrelated then the slope of the regression should be 0.

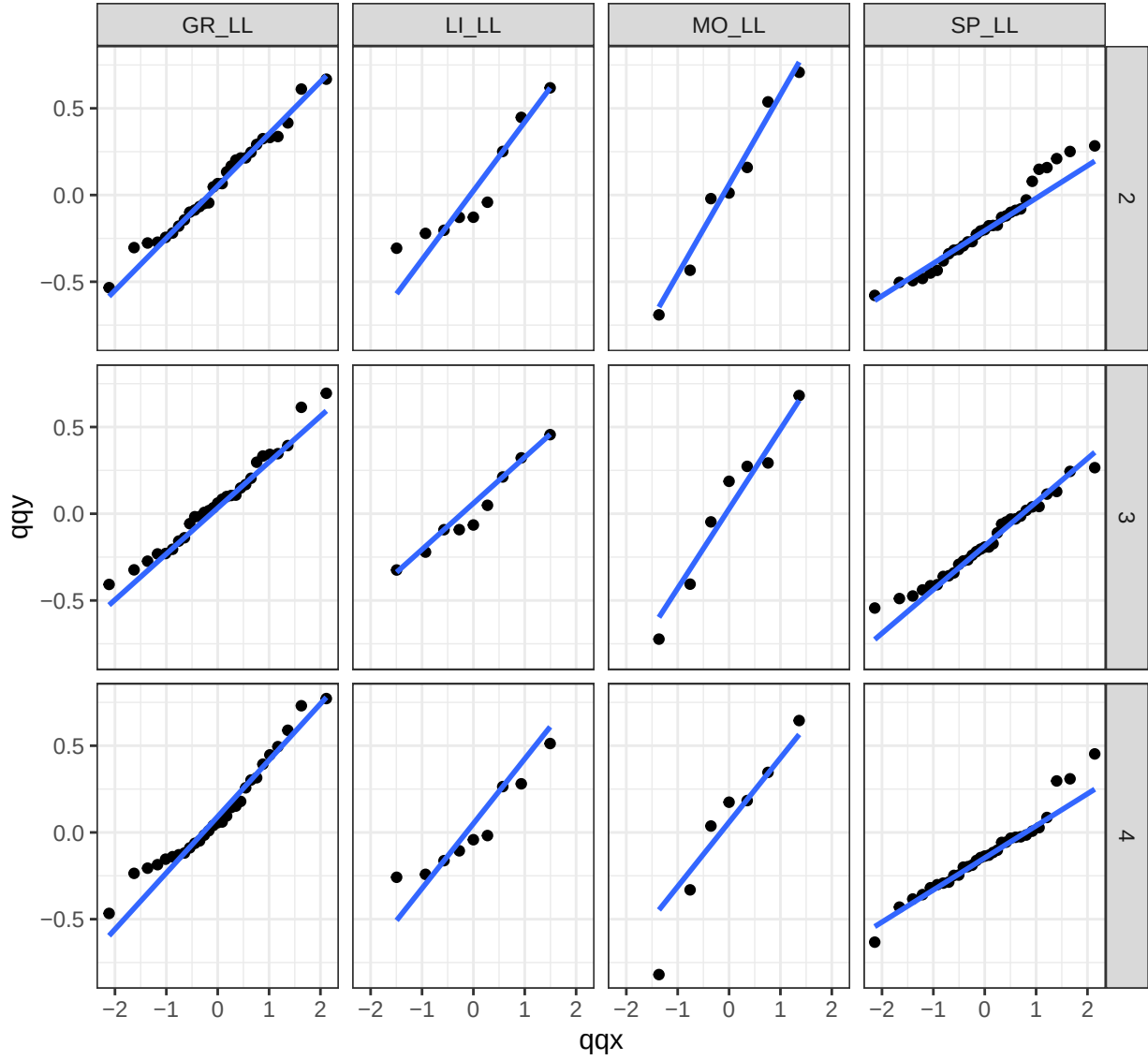


Figure 22. QQ plots to check for log normality (i.e. points and line should coincide) for the discard run.

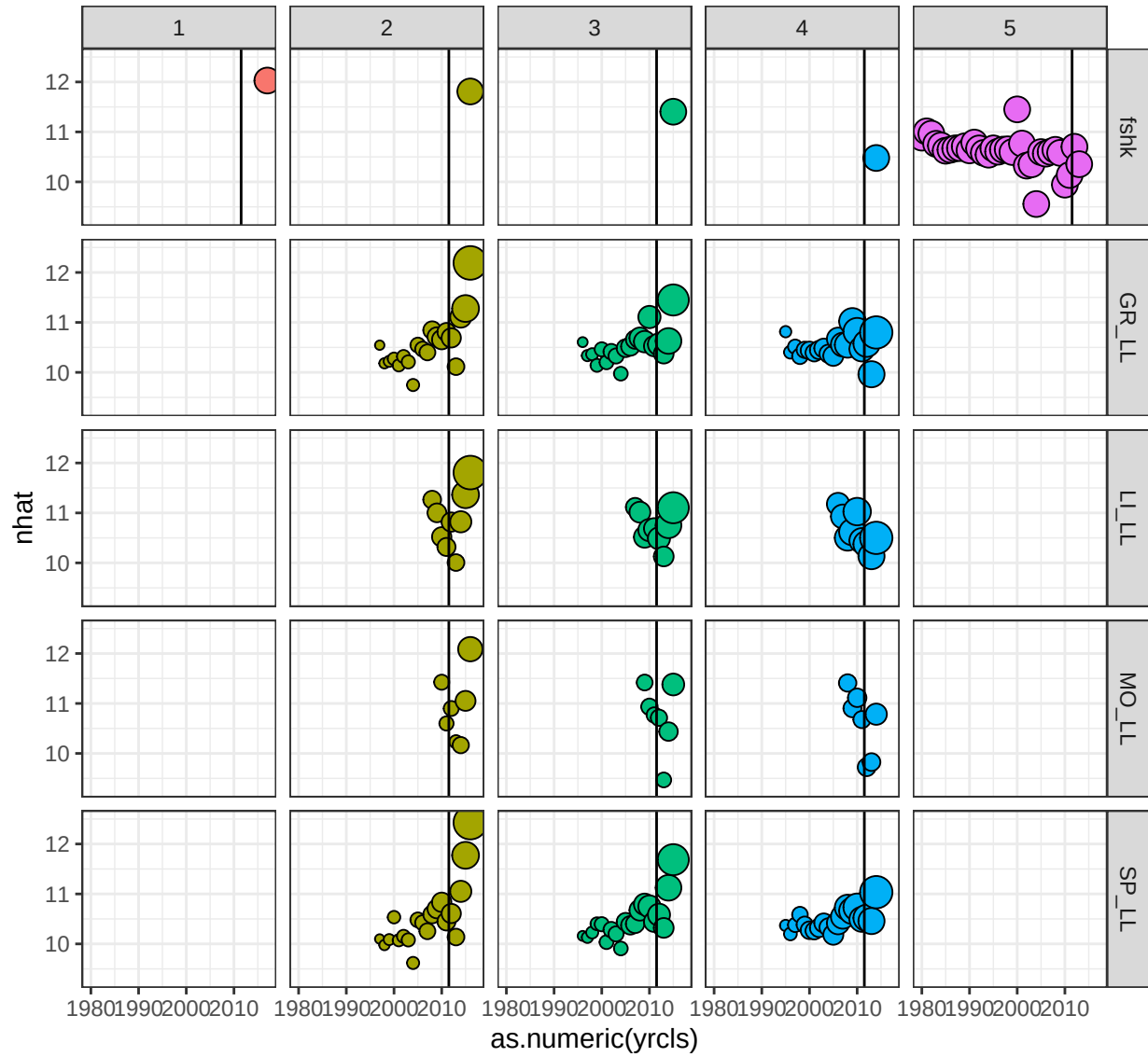


Figure 23. Weights for terminal year Ns for each CPUE observation and shrinkage to the mean $F(\text{fshk})$ for the discard run. Vertical line identifies the range used for shrinkage to age in the terminal ages. Points correspond to individual cohorts.

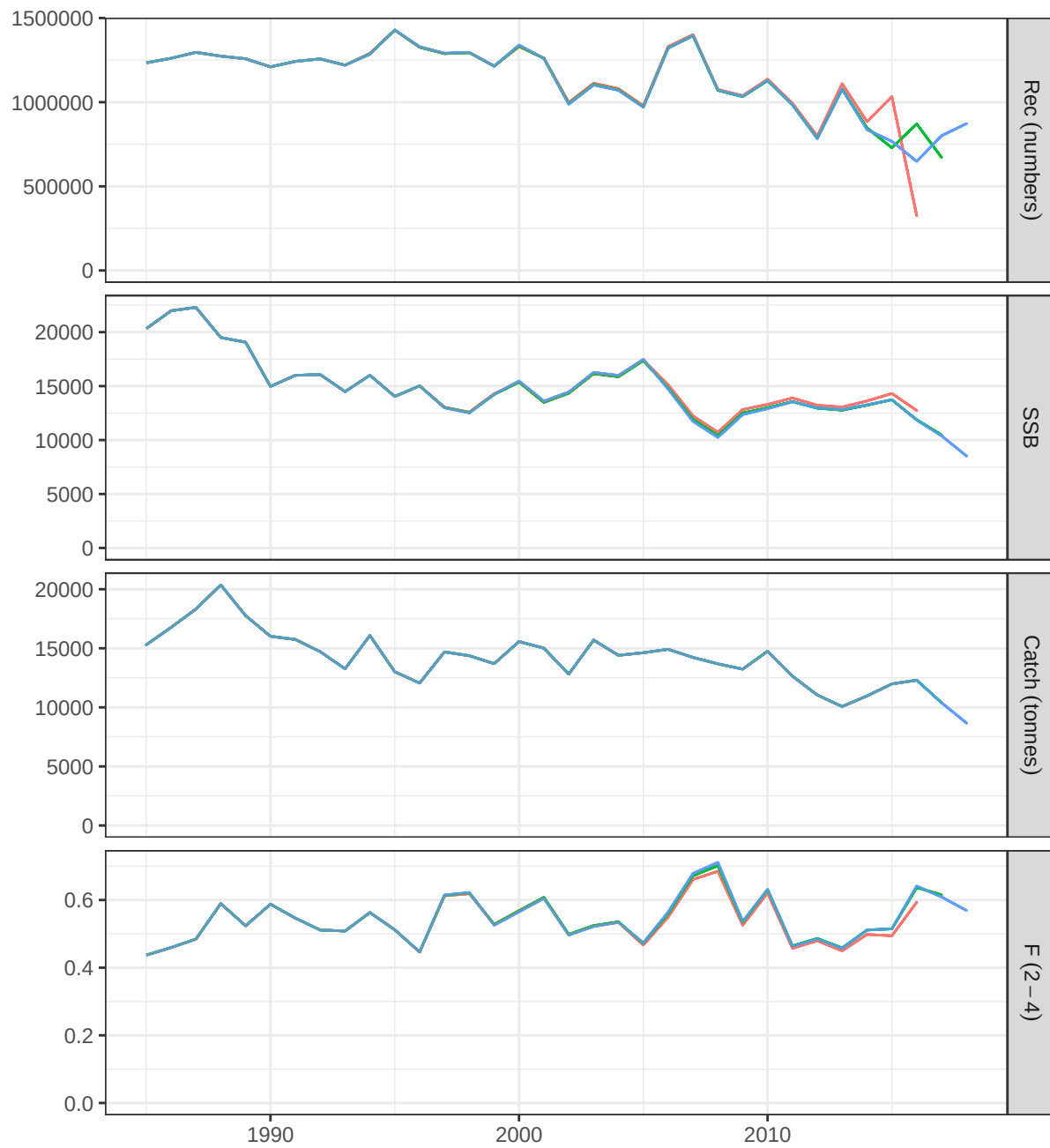


Figure 24. Retrospective plot for the discard run.

f. Comparison among candidate and discard runs

Estimates of the two candidate and the two discard runs are plotted in **Figure 25**.

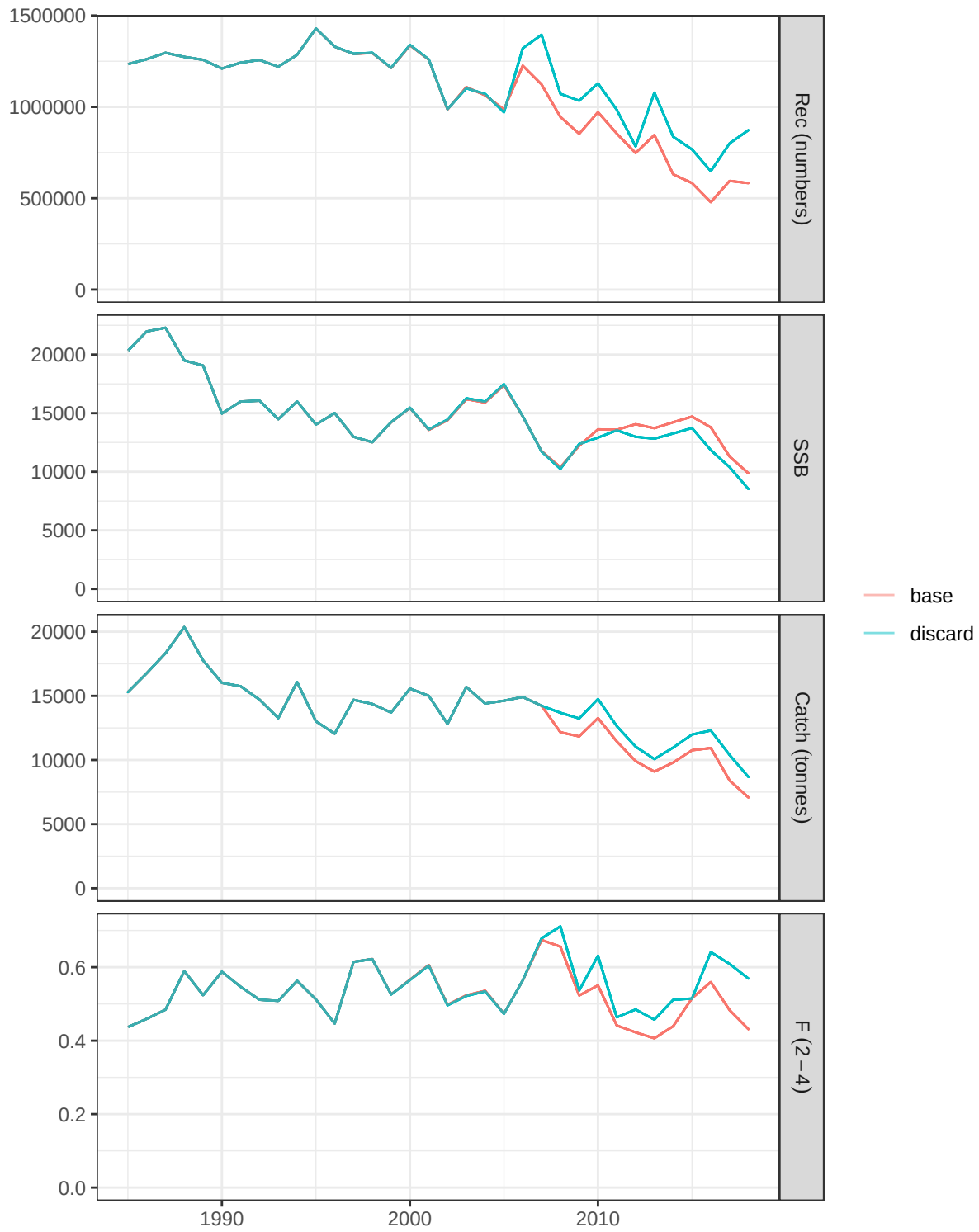


Figure 25. Comparison of estimates among base and discard runs.

c. Biological reference points (BH)

Estimates of the two runs were used to evaluate the state of the stock and obtain biological reference points based upon yield per recruit and stock recruit equilibrium calculations, assuming a Stock-Recruitment Relationship (SRR) of the Beverton and Holt functional form. Given the lack of sufficient data for a proper fit an empirical relationship with high steepness value was developed. **Figures 26-27** illustrate the equilibrium curves, while the estimated reference points are shown in **Table 1**. Results indicate that the stock is overexploited.

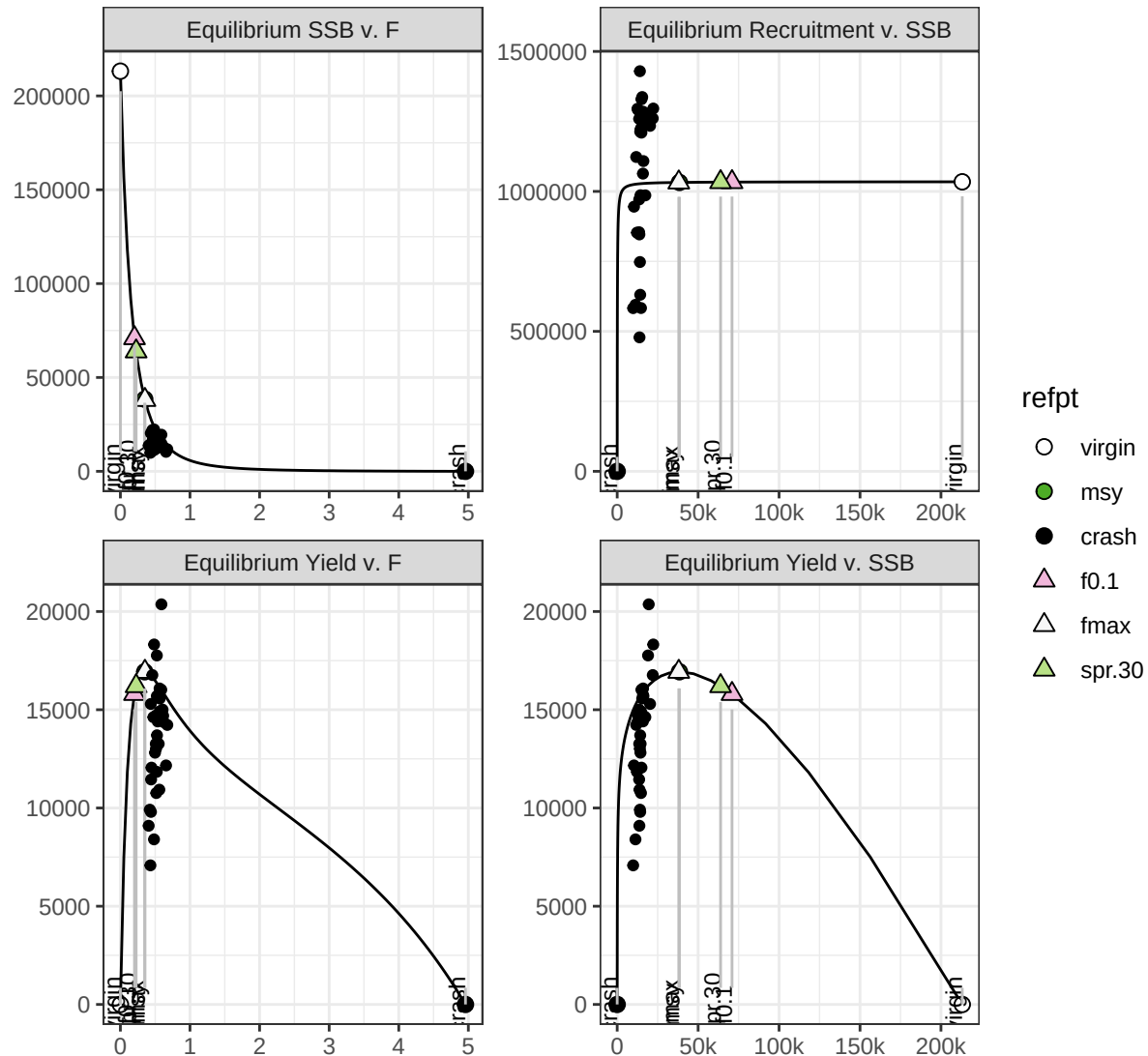


Figure 26. Equilibrium curves with data pairs of the base run assuming a Beverton-Holt SRR.

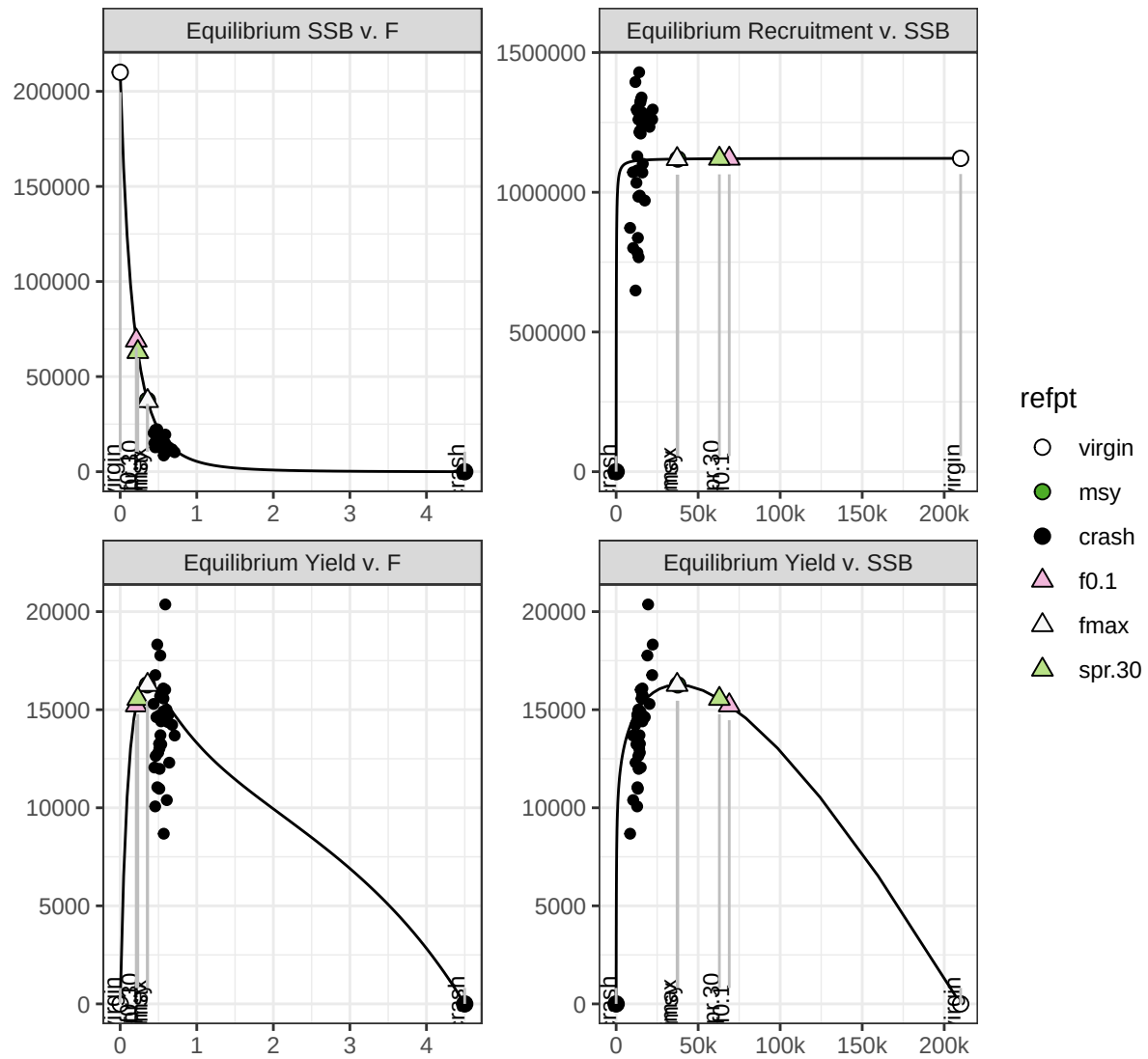


Figure 27. Equilibrium curves with data pairs of the discard run assuming a Beverton-Holt SRR

An object of class "FLPar"

```
      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 1.03e+06 2.13e+05 2.42e+05
    msy   3.50e-01 1.69e+04 1.03e+06 3.84e+04 6.20e+04
    f0.1   2.03e-01 1.58e+04 1.03e+06 7.09e+04 9.64e+04
    spr.30 2.26e-01 1.62e+04 1.03e+06 6.39e+04 8.90e+04
units:  NA
```

An object of class "FLPar"

```
      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 1.12e+06 2.10e+05 2.38e+05
    msy   3.55e-01 1.63e+04 1.12e+06 3.76e+04 6.04e+04
    f0.1   2.10e-01 1.52e+04 1.12e+06 6.89e+04 9.37e+04
    spr.30 2.30e-01 1.56e+04 1.12e+06 6.29e+04 8.74e+04
units:  NA
```

Table 1. Estimates of reference points assuming a Beverton-Holt SRR. From top to bottom: base run, discard run

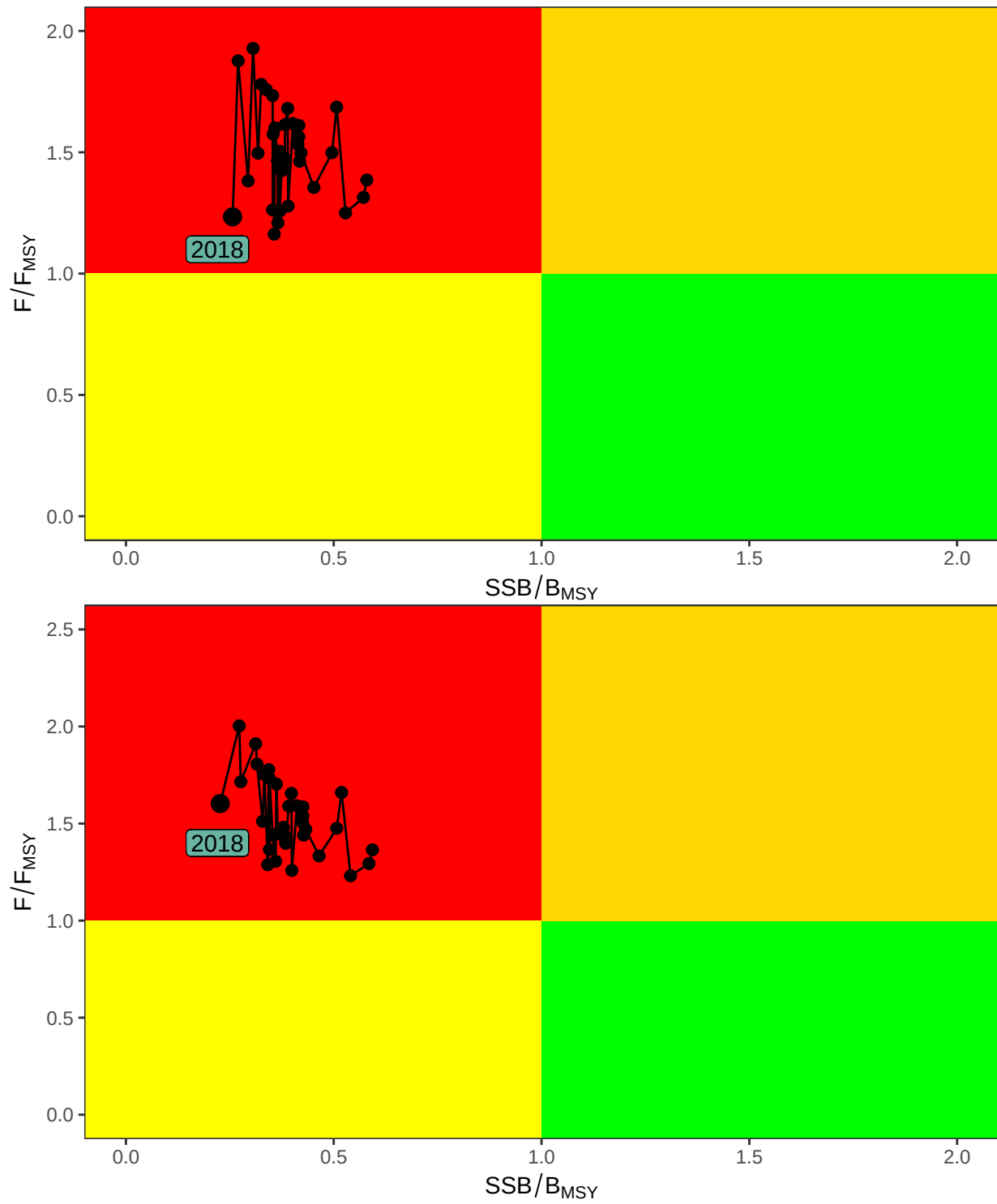


Figure 28. Kobe plots for base (top) and discard (run) assuming a Beverton-Holt SRR.

d. Alternative reference points

Those are based on constant recruitment equal to the geometric mean of the last 10 years.

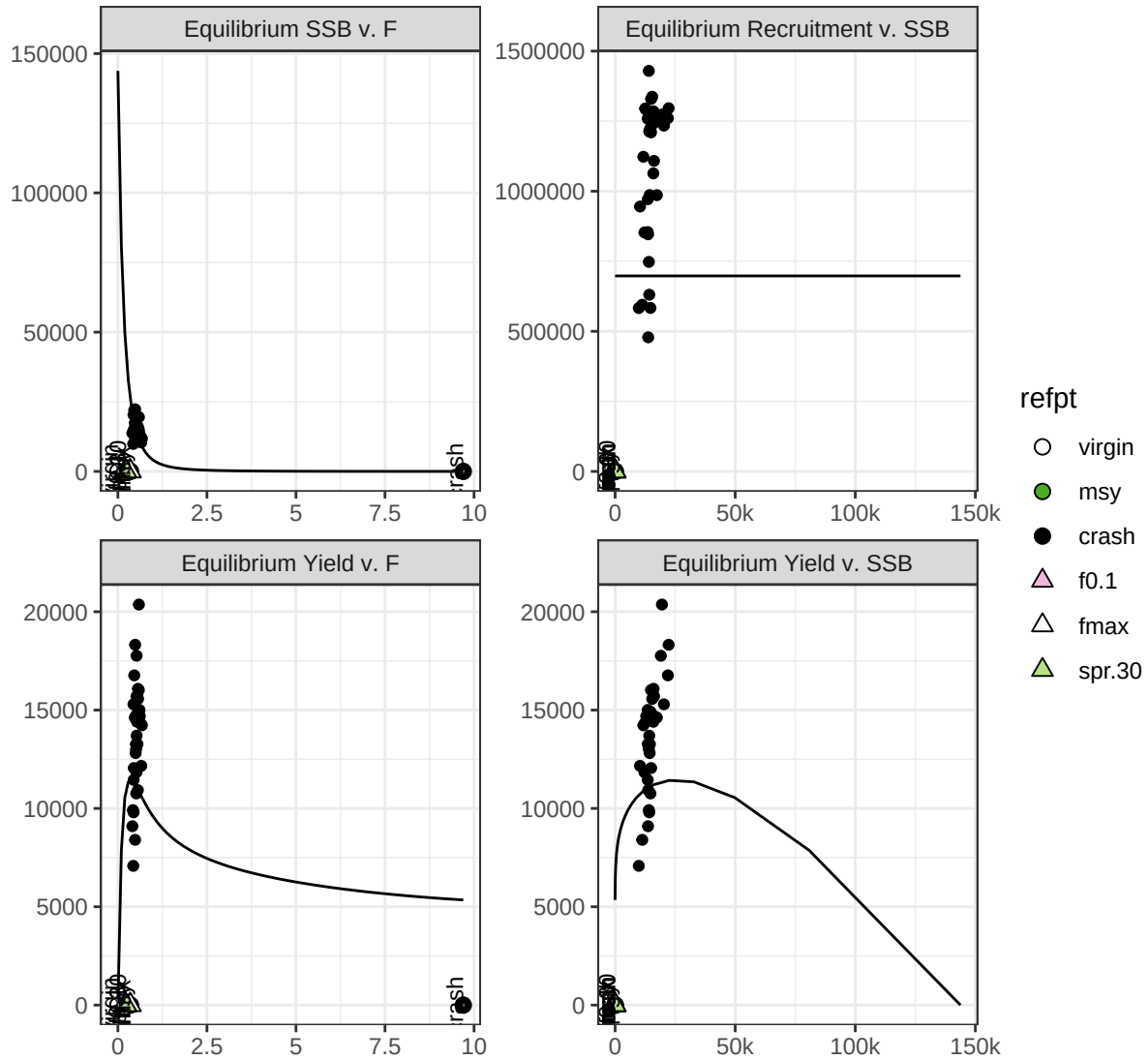


Figure 29. Equilibrium curves with data pairs of the base run assuming constant recruitment.

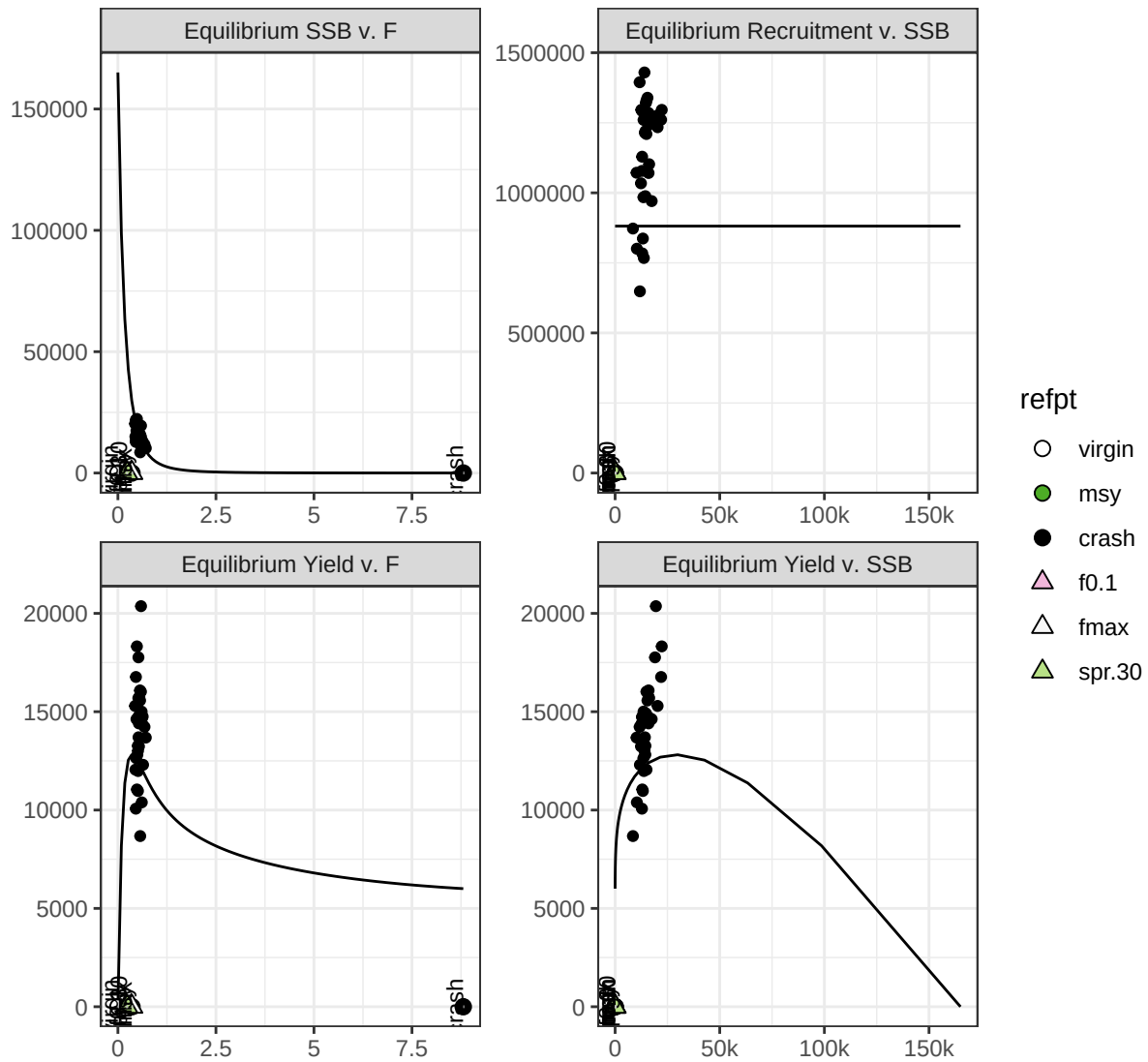


Figure 30. Equilibrium curves with data pairs of the discard run assuming constant recruitment

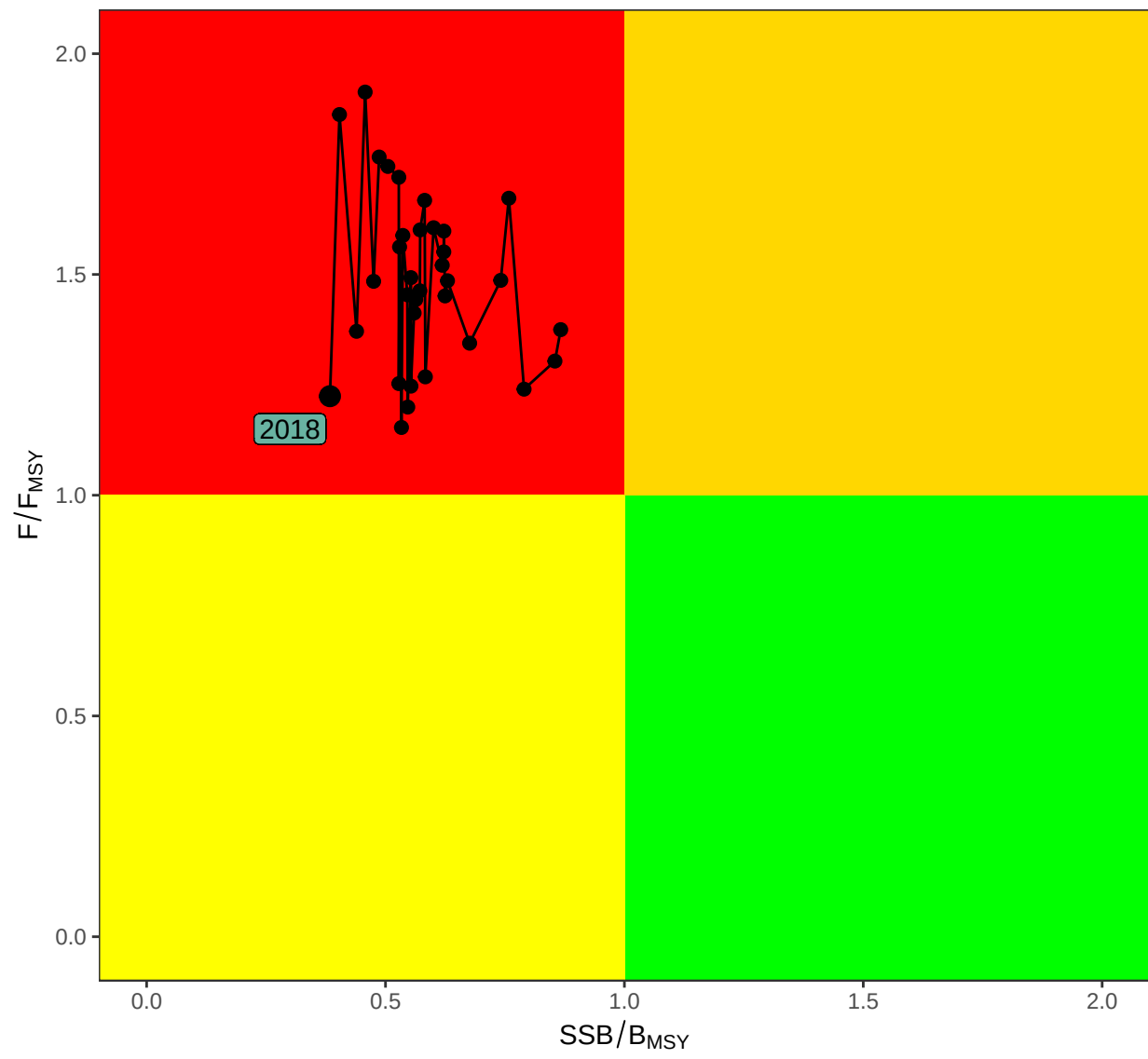
An object of class "FLPar"

```
      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 6.98e+05 1.44e+05 1.63e+05
    msy   3.52e-01 1.14e+04 6.98e+05 2.57e+04 4.16e+04
    f0.1  2.03e-01 1.07e+04 6.98e+05 4.79e+04 6.51e+04
    spr.30 2.26e-01 1.09e+04 6.98e+05 4.31e+04 6.01e+04
units:  NA
```

An object of class "FLPar"

```
      quant
refpt  harvest  yield    rec      ssb      biomass
  virgin 0.00e+00 0.00e+00 8.81e+05 1.65e+05 1.87e+05
    msy   3.58e-01 1.28e+04 8.81e+05 2.93e+04 4.72e+04
    f0.1  2.10e-01 1.20e+04 8.81e+05 5.42e+04 7.37e+04
    spr.30 2.30e-01 1.22e+04 8.81e+05 4.95e+04 6.87e+04
units:  NA
```

Table 2. Estimates of reference points assuming constant recruitment. From top to bottom: base run, discard run



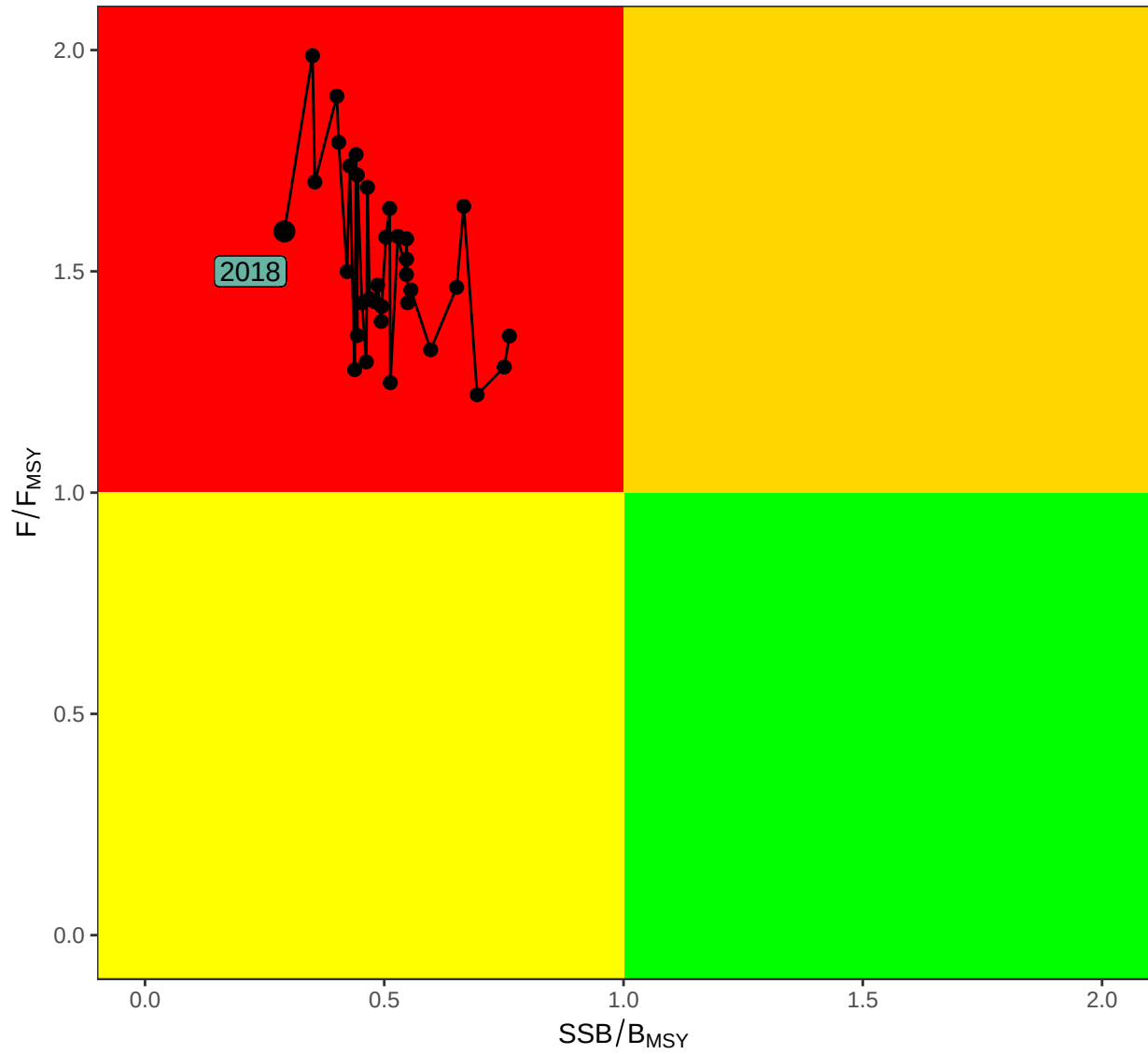


Figure 31. Kobe plots for base (top) and discard (run) assuming constant recruitment.

h. Selection-at-age

Based on the XSA runs, **Figure 28** illustrates the fishery selection pattern. Full selection is achieved at age 3 and it is decreasing afterwards.

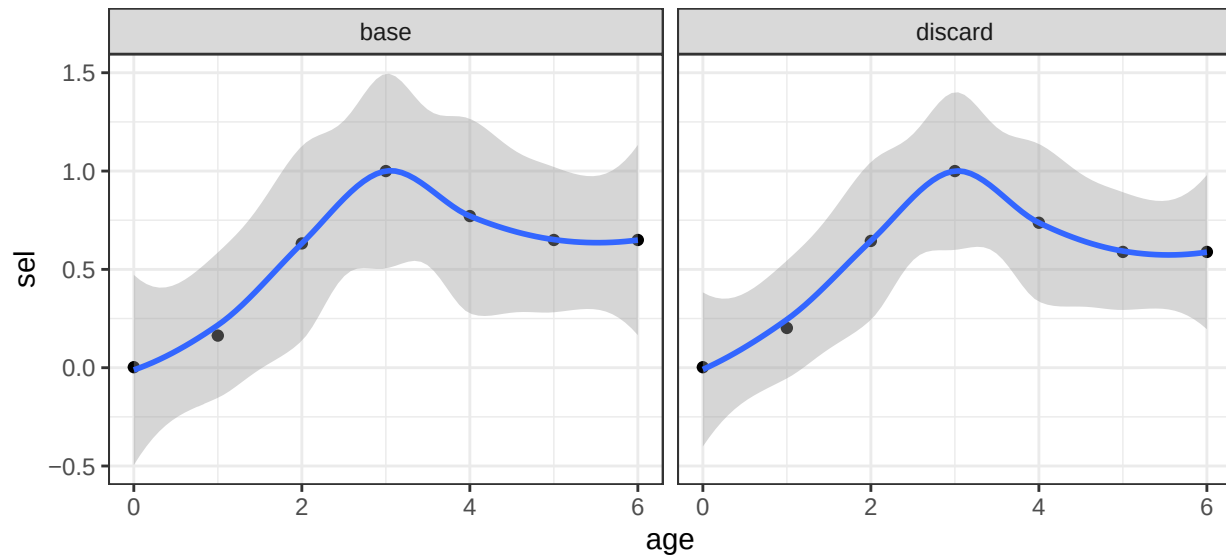


Figure 32. Estimates of selection by age from the two model runs.

References

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